is its lack of security of supply. When a fault occurs on any section of the feeder; a number of consumers will be without supply for a considerable period. The radial system is normally used for rural distribution these days.

10.2.2 The Ring Mains Systems

The system is used most frequently to supply bulk loads such as small industrial loads and medium or large commercial buildings where continuity of supply is of considerable importance. A typical ring mains for feeder and distributors is shown in Fig. 10.4 (*a*).

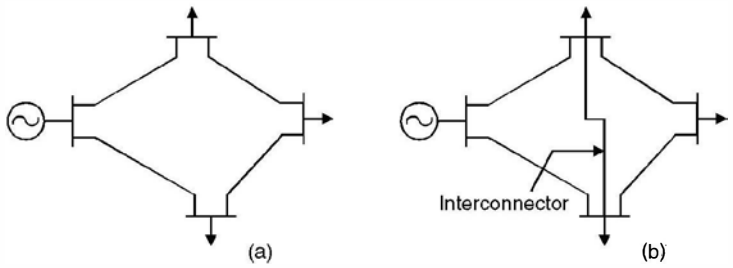


Fig. 10.4. Ring Mains (*a*) Ring (*b*) interconnected.

It can be shown that when an interconnector is used in a ring mains system [Fig. 10.4 (*b*)], it reduces the voltage drop between the points to which it is connected. The ring mains systems is used for urban distribution in contrast to the radial system.

10.3 COST COMPARISON OF SUPPLY SYSTEMS

The various systems considered for cost comparison are

(*i*) d.c. two wires

(*ii*) a.c. single phase two wire

(*iii*) a.c. three phase four wire.

The terms of comparison are (*a*) same maximum phase to ground voltage and (*b*) same efficiency or losses to transmit a given amount of power over a given distance. The quantity to be compared is the volume of copper required in each case.

D.C. two wire with midpoint earthed

Refer to Fig. 10.5 for d.c. two-wire midpoint earthed system. Let V be the voltage between either outer terminal and the earth so that voltage between the two outer terminals is 2V. Let P be the power transmitted. Therefore, the current in each outer conductor will be $I = \frac{P}{2V}$. If R_1 is the resistance of each conductor the copper loss will be

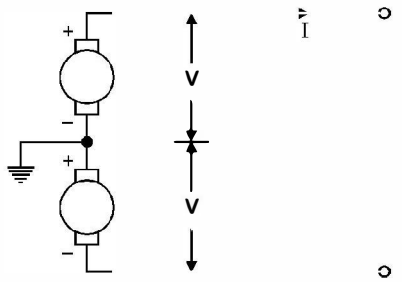


Fig. 10.5. D.C. Two-Wire System.

$$\text{Copper loss} = \frac{2P^2}{4V^2} R_1 = \frac{P^2}{2V^2} R_1$$

A.C. Single Phase Two-Wire

Refer to Fig. 10.6 for a.c. single phase two-wire system with middle point earthed. The maximum voltage between one of the outer wires and the earth is V volts. Therefore,

$$V_{rms} = \frac{V}{\sqrt{2}}$$

The voltage between the two outer conductors is $\frac{2V}{\sqrt{2}} = \sqrt{2}V$

Therefore current in the outer wire $I = \frac{P}{\sqrt{2}\, V \cos \phi}$ if ϕ is the power factor angle of the load.

Now if R_2 is the resistance of each wire, the copper loss = $2 . \frac{P^2}{2V^2 \cos^2 \phi} R_2$

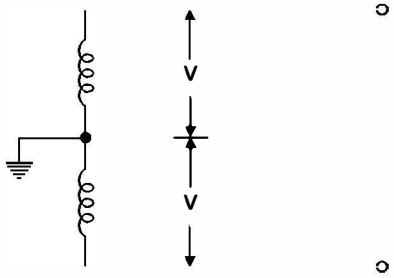


Fig. 10.6. A.C. Two Wire System.

$$\text{Copper loss} = \frac{P^2}{V^2 \cos^2 \phi} R_2$$

A.C. 3-phase 4-wire system Refer to Fig. 10.7 for a 3-phase, 4-wire system

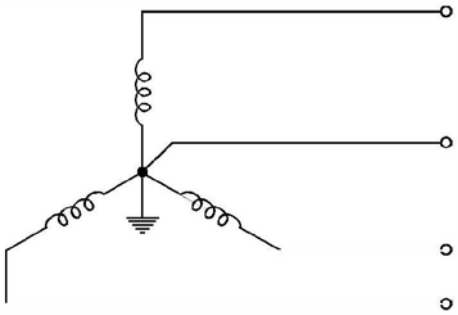


Fig. 10.7. Three Phase 4-Wire System.

The maximum voltage between one phase and grounded neutral is V volts.

$$V_{rms} = \frac{V}{\sqrt{2}}$$

$\therefore$ Current per phase under balanced condition = $\frac{P}{3 \times \frac{V}{\sqrt{2}} \cos \phi}$

where ϕ is the power factor angle of load. If R_3 is the resistance per conductor, copper loss is given by

$$\text{Copper loss} = 3\,\frac{2P^2}{9\,V^2\cos^2\phi}\,R_3 = \frac{2}{3}\,\frac{P^2}{V^2\cos^2\phi}\,R_3$$

Equating the copper loss due to 3 phase 3-wires as calculated above with d.c. two wire system

$$\frac{P^2 R_1}{2V^2} = \frac{2P^2}{3\,V^2\cos^2\phi}\,R_3$$

or
$$\frac{R_1}{R_3} = \frac{4}{3}\times\frac{1}{\cos^2\phi} \quad \text{or} \quad \frac{a_3}{a_1} = \frac{4}{3}\,\frac{1}{\cos^2\phi}$$

Since in case of three-phase three-wire system there are three wires and in case of d.c. two-wire system there are two wires, the volumes required for the same length l, are $3a_3\,l$ and $2a_1 l$ respectively

$$\frac{3a_3 l}{2a_1 l} = \frac{3}{2}\,\frac{a_3}{a_1} = \frac{3}{2}\,\frac{4}{3}\,\frac{1}{\cos^2\theta}$$

In case the fourth wire, in case of 3-phase, 4-wire system is of the same size as the phase wires, the volume of copper required will be $4a_3 l$ and the ratio

$$\frac{4a_3 l}{2a_1 l} = \frac{2a_3}{a_1} = 2\,\frac{4}{3}\,\frac{1}{\cos^2\theta} = \frac{8}{3}\,\frac{1}{\cos^2\theta}$$

Similarly, to compare the volume of copper required in case of a.c. single phase two wire system with the d.c. two-wire system we equate copper losses

$$\frac{P^2}{2V^2}\,R_1 = \frac{P^2}{V^2\cos^2\phi}\,R_2$$

or
$$\frac{R_1}{R_2} = \frac{2}{\cos^2\phi} \quad \text{or} \quad \frac{a_2}{a_1} = \frac{2}{\cos^2\phi}$$

Since the number of wires required is two in each case the ratio of copper volume is

$$1 : \frac{2}{\cos^2\phi}$$

Therefore, the ratio of copper volume required in the three cases under study when d.c. two wire system is taken as reference are

$$1 : \frac{2}{\cos^2\phi} : \frac{8}{3}\cdot\frac{1}{\cos^2\phi}$$

In case the system is single phase two wire with neutral grounded the reader will find that the ratio of volume of copper with reference to d.c. two wire will be

$$1 : \frac{8}{\cos^2\phi}$$

Example 10.1. *A d.c. two-wire system is to be converted into a.c. 3-phase 3-wire system by the addition of a third conductor of the same cross-section as the existing conductors. If the rms voltage between wires and the percentage loss in the line remains unchanged, determine the additional load which can now be supplied. Assume balanced load of unity p.f.*

Solution. (*a*) d.c. two-wire : Let the power transmitted in this case be P and V be the voltage between wires

$$I = \frac{P}{V}$$

Therefore $$\text{Loss} = \frac{2P^2}{V^2} R$$

where R is the resistance of each wire

Therefore,

$$\text{Per unit loss} = \frac{2P}{V^2} R$$

(*b*) 3-phase 3-wire : The r.m.s. voltage = V volts.

Let P' be the power transmitted then $I' = \frac{P'}{\sqrt{3}V}$ for unity p.f.

Therefore,

$$\text{Copper loss} = 3\left(\frac{P'}{\sqrt{3}V}\right)^2 R = \frac{P'^2}{V^2} R$$

Therefore $$\text{per unit loss} = \frac{P'}{V^2} R$$

Equating the per unit loss we have

$$\frac{2P}{V^2} R = \frac{P'}{V^2} R \quad \text{or} \quad P' = 2P$$

This shows that 100% additional power can be supplied by 3-phase 3-wire system.

10.4 POWER FACTOR IMPROVEMENT

Power factor of a circuit or installation is defined as the ratio of power consumed by the circuit (in watts or kW or MW) to the total complex power input to the circuit or the installation and is denoted by cosϕ where ϕ is the power factor angle and is the angle between the circuit voltage and current through the circuit. Consider Fig. 10.8 (*a*) where the current lags

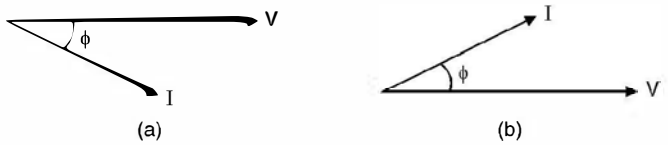


Fig. 10.8. Phasor diagram (*a*) Lagging (*b*) Leading.

the voltage by an angle ϕ whereas in (*b*) it leads the voltage by an angle ϕ. The active power consumed in both the circuit is $|V|\ |I|\ \cos\phi$ and the complex power is $|V|\ |I|$. The units are watts and volt-amperes respectively. However, the reactive power in (*a*) is being absorbed by the circuit as it is an inductive circuit whereas in (*b*) the reactive power is being supplied (lagging) or leading reactive power is taken from the supply due to the presence of capacitor in circuit (*b*) The p.f. in a circuit is decided by what the net reactive power is present in the circuit. So we first

study the sources and sinks of reactive power. In brief the inductance wherever it be, whether the winding of a generator or a motor or the series inductance of a line or a simple coil absorb (sink) the reactive power whereas the capacitance of a line or static capacitor or synchronous motor when overexcited acts as a synchronous capacitor.

In a circuit if both the inductor and capacitor are present, the reactive power due to inductor (phasor) is in phase opposition to the reactive power due to capacitor. The net reactive power will decide whether the circuit as a whole is inductive or capacitive. However, if the two reactive powers are equal in the circuit, the net reactive power requirement of the circuit is zero and hence the p.f. of the circuit is unity. The three possible situations are shown in Fig. 10.9. Conventionally, the reactive power due to inductor is taken as positive and that due to capacitor as negative. This is because the reactive power supplied by the source is taken as positive and that supplied by the load to the source is taken as negative. This convention is used in Fig. 10.9.

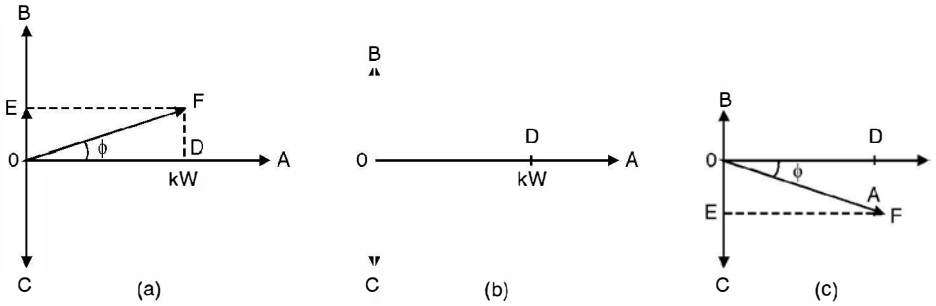


Fig. 10.9. Effect of reactive power on p.f.

In Fig. 10.9 (*a*) the inductive reactive power is greater than the capacitive and hence $OB - OC(BE) = OE$ is the net positive reactive power and OF the complex power in K VA and hence p.f.

$$\cos \phi = \frac{OD}{OF} \text{ lagging}$$

In Fig. 10.9 (*b*) since $OB = OC$ hence the net reactive power is zero and the p.f. of the circuit is unity *i.e.* the circuit behaves as resistive. In Fig. 10.9 (*c*) $OC > OB$, hence the net capacitive reactive power is $OE = OC - OB\,(CE)$ and the power factor is $\cos \phi = \frac{OD}{OF}$ lead. Reactive power in a circuit is defined as the wattless component of the complex power *i.e.* the component which does not do any effective work yet it is required in most of the electrical equipments and is sort of necessary evil just as friction is a necessary evil in our daily life activities. Reactive power is the time rate of change of electromagnetic energy stored in an inductor or the electrostatic energy stored in a capacitor. The presence of reactive power not only affects the p.f. of the circuit, it also affects the voltage levels at different points of the circuit. It is found that in a circuit if reactive power generated is more than the reactive power absorbed by the circuit, the voltage levels rise and voltage levels dip otherwise.

In an electric power system it is difficult to predict the load variation accurately. The load devices may vary from a few watt night lamps to multi megawatt induction motors. The following categories of loads are present in a system.

(*i*) Motor devices 70%
(*ii*) Heating and lighting equipment 25%
(*iii*) Electronic equipment 5%

10.4.1 Causes of Low p.f.

Among the chief causes of poor p.f. may be mentioned, induction motors (which form almost 70% of the total load) particularly in small sizes (fractional hp motors) and large motors when running partially loaded, arc and induction furnaces, electric discharge lamps (associated with ballasts) and transformers. The transformers are also predominantly inductive circuits and in fact each megawatt generated, transmitted and distributed requires an equivalent of 3 to 4 MVA capacity transformer, at different voltage levels.

10.4.2 Methods to Improve p.f.

Following methods can be used for improvement of p.f. of a circuit.

(*i*) Use of high p.f. machines
(*ii*) Use of static capacitors
(*iii*) Use of synchronous capacitors.

(*i*) **Use of high p.f. machines.** There are several types of especially designed motors which operate at a high p.f. of approximately unity (between 0.95 to 0.97 pf) notably the synchronous motor and the compensated induction motors. In general, it is found that such motors are more expensive and have a higher maintenance cost than the plain induction motors, so that in choosing such motors these additional costs must be balanced against the savings in energy cost due to their higher p.f. Of the above types, the synchronous motor is particularly useful as it can be very easily designed to operate at a leading p.f. and, therefore, compensate for the lagging p.f. of other machines.

(*ii*) **Use of static capacitor.** Consider Fig. 10.10 where load (inductive) is connected to the source.

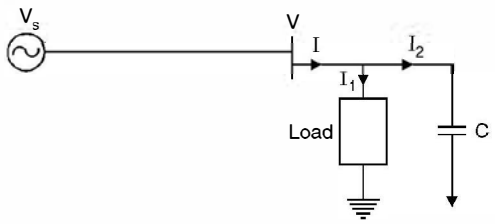


Fig. 10.10. Use of Static Capacitor.

For improvement of p.f. or improvement of voltage profile it is always desirable to connect a capacitor across the inductive load as shown in Fig. 10.10. As explained in article 7.7, a capacitor takes leading vars or it is equivalent to saying that it supplies lagging vars. Consider the phasor diagram in Fig. 10.11.

In Fig. 10.11 (*a*) the phasor diagram represents the kW, kVAr and kVA without the capacitor where the p.f. angle is ϕ and p.f. cos ϕ.

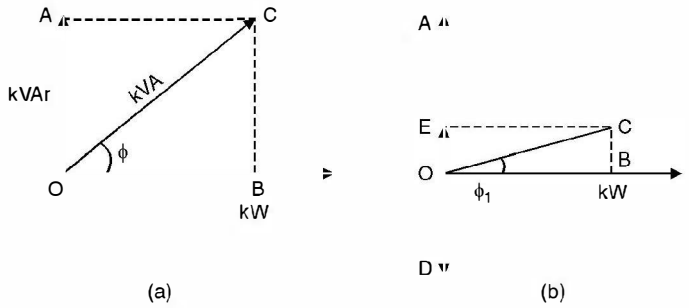


Fig. 10.11. p.f. improvement using static capacitor.

In Fig. 10.11 (*b*) when a capacitor *C* is connected across the load, it takes leading VArs or supplies the lagging VArs to the load and hence, the net lagging VArs to be supplied by the source are reduced by an amount *OD* and the source has to supply kVArs equivalent to $OA - OD = EO$ and the p.f. of the busbar improves to $\cos \phi_1$ as $\phi_1 < \phi$. It is to be noted very carefully that the p.f. of the load does not change here, it is a fixed quantity but it is the p.f. of the system which is improved with the use of a suitable value of capacitor across the load.

(*iii*) **Use of synchronous capacitor.** This has already been discussed at length in Article 7.7. The readers are advised to go through this article.

However, it may be mentioned here that to improve the p.f. above 0.95 requires a large amount of leading kVAr and the cost of the equipment to improve it to such high values may well out weigh any savings in energy costs.

Example 10.2. *A slow speed alternator works in parallel with a turbo-alternator, the combined output being 2500 kW at 0.8 p.f. lag. If the turbo altenator provides 1000 kW at unity p.f., at what p.f. will the slow speed machine work ?*

Solution. Total active power of the two alternators 2500 kW

Total reactive power $2500 \tan \cos^{-1} 0.8$

$$= 2500 \times 0.75$$

$$= 1875 \text{ kVAr}$$

Active power supplied by turbo alternator 1000 kW

Reactive power supplied by turbo alternator = zero

Hence active power supplied by the slow speed alternator = 2500 – 1000 = 1500 kW

And reactive power supplied by the slow speed alternator = 1875 kVAr

Hence p.f. of the slow speed alternator $\cos \tan^{-1} \dfrac{1875}{1500} = 0.624$ **Ans.**

Example 10.3. *Two alternators working in parallel supply a lighting load of 3000 kW and a motor load aggregating 5000 kW at a p.f. of 0.71. One machine is loaded to 5000 kW at p.f. 0.8 lag. Determine the load and p.f. of the other machine.*

Solution. Total load active power requirement

Lighting load	3000 kW
Motor load	5000 kW
Total	8000 kW

Total reactive power requirement of the load

Lighting load = zero

Motor load $= 5000 \times \tan \cos^{-1} 0.71$

$= 5000 \times 0.9918$

$= 4959$ kVAr

Active power supplied by one machine = 5000 kW

Reactive power supplied by one machine $= 5000 \tan \cos^{-1} 0.8$

$= 5000 \times \frac{3}{4}$

$= 3750$ kVAr.

Active power supplied by the other machine = 8000 – 5000 = 3000 kW

Reactive power supplied by the other machine = 4959 – 3750 = 1209 KVAr

p.f. of the other machine $\cos \tan^{-1} \frac{1209}{3000} = 1209$ kVAr

$= 0.927$ **Ans.**

Example 10.4. *A single phase motor takes a current of 40A at p.f. 0.7 lag from a 440 V 3-phase 50 Hz supply. Determine the value of a shunt capacitor so as to raise the p.f. to 0.9 lag, the load remaining the same.*

Solution. The power active taken by the motor

$$\frac{440}{\sqrt{3}} \times 40 \times 0.7 = 7112 \text{ watt}$$

and reactive power $= \frac{440}{\sqrt{3}} \times 40 \sin \cos^{-1} 0.7$

$= 254 \times 40 \times 0.714$

$= 7254$ VArs.

The new p.f. is 0.9 lag, power remaining same reactive power requirement for this p.f.

$7112 \tan \cos^{-1} 0.9 = 3444$ VArs

The VArs to be supplied by the shunt capacitor are 7254 – 3444 = 3810 VArs

The value of capacitor is given by

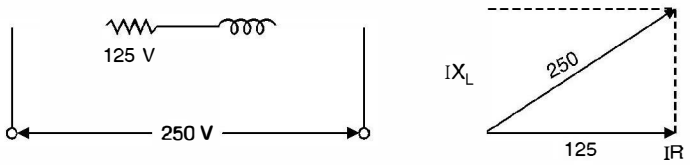
$$V^2 wc = 254^2 \times 314\ c = 3810$$

or $$c = \frac{3810}{254^2 \times 314} = 188\ \mu\text{F}$$ **Ans.**

Example 10.5. *A 500 W discharge lamp takes a current of 4A at unity p.f. Calculate the inductance of a choke required to enable the lamp to work on 250 V 50 Hz mains. Determine also the capacitance to be connected across the mains to bring the resultant p.f. to unity.*

Solution. Since the lamp takes 4A at unity p.f. the supply voltage should be $\frac{500}{4} = 125$ volts.

However we want to connect, the same lamp across 250 volt. The circuit is as follows



From the phasor diagram, the voltage across the choke would be

$$\sqrt{250^2 - 125^2} = 216.5 \text{ volt}$$

The reactance of the choke $= \frac{216.5}{4} = 54.12$

$$wL = 54.12$$

$$314\ L = 54.12. \quad \text{Hence } L = 0.172 \text{ H}$$

The reactive power requirement of the choke $I^2wL = 16 \times 0.172\, w$

and if C is the capacitance required across the supply, the reactive power supplied by the capacitor is

$$V^2wC = 250^2wC = 16 \times 0.172\, w$$

or $$250^2C = 16 \times 0.172$$

or $$C = 44\ \mu\text{F}. \quad \textbf{Ans.}$$

10.5 GENERATION—NON-CONVENTIONAL SOURCES

The contemporary non-conventional sources of energy like wind, tidal, solar etc. were the conventional sources until James Watt invented the steam engine in the eighteenth century. In fact, the New World was explored by man using wind-powered ships only. The non-conventional sources are available free of cost, are pollution-free and inexhaustible. Man has used these sources for many centuries in propelling ships, driving windmills for grinding corn and pumping water, etc. Because of the poor technologies then existing, the cost of harnessing energy from these sources was quite high. Also because of uncertainty of period of availability and the difficulty of transporting this form of energy, to the place of its use are some of the factors which came in the way of its adoption or development. The use of fossil fuels and nuclear energy replaced totally the non-conventional methods because of inherent advantages of transportation and certainty of availability; however these have polluted the atmosphere to a great extent. In fact, it is feared that nuclear energy may prove to be quite hazardous in case it is not properly controlled.

In 1973 the Arab nations placed an embargo on petroleum. People began to realise that the fossil fuels are not going to last longer and that remaining reserves should be conserved for the petro-chemical industry. But unfortunately, both nuclear and coal energy pose serious environmental problems. The combustion of coal may upset the planet's heat balance. The production of carbondioxide and sulphurdioxide may adversely affect the ability of the planet to produce food for its people. Coal is also a valuable petro-chemical and from long term point of view it is undesirable to burn coal for generation of electricity. The major difficulty with nuclear energy is waste disposal and accidental leakage (e.g. leakage at Chernobyl nuclear power plant).

As a result of these problems, it was decided by almost all the countries to develop and harness the non-conventional sources of energy, even though they are relatively costlier as compared to fossil-fuel sources. It is hoped that with advancement in technology and more and more research in the field of development of non-conventional sources of energy, these sources may prove to be cost-effective as well. The future of wind, solar, tidal and other energy sources is bright and these will play an important role in the world energy scenario.

The following sections have been devoted to the study of some of the important non-conventional sources of energy.

10.5.1 Tidal Power

Tidal or lunar energy as it is sometimes called, has been known to mankind since time immemorial. Various devices, particularly the mills were operated using tidal power. In the past water supply of London was pumped to a water tower by a mill operated by the tidal power (which consisted of a large paddle wheel mounted on a raft and fastened between two of the piers of old London Bridge). The tidal power has been used to irrigate fields in Germany and to saw firewood in Canada.

Tides are caused by the combined gravitational forces of Sun and Moon on the waters of the revolving Earth. When the gravitational forces due to the Sun and the Moon add together, tides of maximum range, called spring tides, are obtained. On the other hand, when the two forces oppose each other, tides of minimum range, called neap tides, are obtained. In one year there are approximately 705 full tidal cycles.

Even though many problems have to be overcome in tidal power development, this form of power has certain definite advantages. Output of a tidal power station is independent of the seasonal changes and can be predicted well in advance, as it depends on the cosmic phenomenon. It is possible to predict the amount of power and the time at which it will be available throughout the year. This power can, therefore, be utilized at the proper position of the load curve.

More than fifty sites have been identified in the world for possible generation of tidal power. As more and more technological advancement take place, even more sites could be identified for tidal power development. Some of the important sites are :

(*i*) La Rance (France), (*ii*) Severn Barrage (UK), (*iii*) White Sea (USSR), (*iv*) Passamaquoddy (USA), (*v*) Gulf of Cambey (India) and (*vi*) Gulf of Kutch (India).

The maximum tidal range in the Gulf of Cambey is about 10.8 m and is quite attractive for a tidal plant. However, the silt charge of the Gulf of Cambey is relatively high and needs a closer study for further development.

The Gulf of Kutch has a maximum spring tide of 7.5 m and the silt charge is relatively low.

10.5.2 Wind Power

The wind wheel, like the water wheel, has been used by man for a long time for grinding corn and pumping water. Ancient seamen used wind power to sail their ships. With the development of the fossil fuelled and hydro-electric plants, there was decline in the use of wind power due to the less cost involved in the new methods. Another difficulty with wind power was the problem of energy storage. The energy could not be made available, on demands, due to uncertainties of wind. Due to these two reasons, no further attempt was made to develop wind power for large scale power generation.

In recent years, however, as a result of energy crisis in the world, it has been decided to investigate all possible means of developing power, as alternatives to fuel fired plants. The wind could supply a significant portion of the world's energy demand. An estimate by an American Professor indicates the potentialities of wind power. According to him about 350,000 wind mills each rated for about 1250 KW to 2200 KW could develop power of the order of 190,000 MW. With the advancement in the knowledge of aero-dynamics it has been possible to build larger and more efficient wind power plants. A typical example is the 1250 KW installation at Grandpa's Knol in U.S.A. Whereas some success has been achieved in developing small and medium size plants, the prospects of large scale generation *i.e.* 1 MW or above are not, as yet very encouraging.

Economic development of wind power requires selection of sites where high specific outputs are compatible with reasonable cost of construction of plant. It is, therefore, necessary to obtain wind velocity duration curve for a particular site and to know the output of the machine for varying wind velocities. The maximum efficiency of the wind power plant is found not to exceed 40%.

The wind power plants can be operated in combination with steam or hydro power station, which will lead to saving in fuel and increase in firm capacity, respectively of these plants.

Wind energy can prove to be a potential source of energy for solving the energy problem. It can certainly go a long way to supply pollution-free energy to millions of people, living in the villages all over the world.

The economic viability of wind mills is better in situations where conventional transmission costs are extremely high (because of inaccessibility and small load) or where continuous availability of supply is not essential so that only a limited amount of storage or standby power need be provided.

10.5.3 Geothermal Power

Many geothermal power plants are operating throughout the world. Although larger geothermal power plants are in operation in America today, it is to the credit of the Italians that the first impressive breakthrough in geothermal power exploitation was achieved. The oldest geothermal power station is near Larderello in Italy, which has an installed capacity of 380 MW. In Newzealand geothermal power accounts for 40% of the total installed capacity, whereas in Italy it accounts for 6%.

It is a common knowledge that the earth's interior is made of a hot fluid called 'magma'. The outer crust of the earth has an average thickness of 32 Km and below that, is the magma. The average increase in temperature with depth of the earth is 1°C for every 35 to 40 metre depth. At a depth of 3 to 4 Kms, water boils up and at a depth of about 15 Kms, the temperature is, in the range of 1000° to 1200°C. If the magma finds its way through the weak spots of the earth's crust, it results into a volcano. At times, due to certain reasons the surface water penetrates into the crust, where it turns into steam, due to intense heat, and comes out in the form of springs or geysers. Moreover, the molten magma also contains water, which it releases in the form of steam, which could be utilized for electric power generation.

Combined Operation of Geothermal Plant

It is well known that a composite power system can be supplied more economically by a combination of two main types of plants :

(*i*) Base Load Plant which is characterised by high fixed cost and low variable cost.

(*ii*) Peak Load Plant which is characterised by low fixed cost and high variable cost.

In case of a geothermal plant, the usual practice is to regard all the production cost as fixed cost, with zero variable cost as no fuel is required for its operation. This is justified by the fact, that once geothermal steam has been made available by means of capital spent on exploitation, drilling and pipe work, it may be regarded as free. Geothermal plants are, therefore, ideally rated as base load plants. Most of the plants today are being used as base load plants as they can achieve annual plant load factor of 90% or more—higher than that obtainable from thermal or nuclear plant.

The commercial viability of geothermal power plant as compared to other sources, depends upon the cost of alternative power sources and other local factors. As a rule of thumb, the following guidelines may be followed to assess its viability :

(*i*) The fluid temperature at the bottom of the bore should be at least 180°C.

(*ii*) A temperature of 180°C should be available at depths not exceeding 3 Kms.

(*iii*) The yield from a $24\frac{1}{2}$ cm bore should be a least 20 tons/hr of steam.

The following are some of the geothermal power projects in operation :

Larderello	:	Italy
Geysers	:	California (USA)
Cerro Prieto	:	Mexico
Hatchobaru	:	Japan
Matsukawa	:	Japan
Paratunka	:	USSR
Wairakei	:	New Zealand
Pugga Multipurpose Project (Ladakh)	:	India

At present geothermal energy makes a very small, but locally important, contribution to world energy requirements. This situation will not change unless important technological advances are made. Environmentally, it is probably the least objectionable form of power generation available at present, with, the exception of hydroelectric methods.

10.5.4 Wave Power

Another source of non-conventional energy generation is the wave power. The major problem with the wave power is that it is not concentrated at a place. If means could be developed for collecting the energy in the wave, spread over a large surface area, and concentrating it into a relatively small volume, the prospects, would be considerably improved.

It has been observed that a typical wave measures 2 to 3 metres in height throughout the year. The energy per square metre of wave surface area is given as $1/2\,\rho\, ga^2$ where ρ is density of sea water, g is acceleration due to gravity and a is the amplitude of the wave. In the Atlantic, the wave period T is around 9 s, and the average velocity of propagation of wave is 14 m/s. It has been observed that a power flow of around 70 KW for every metre of wave front, can be obtained. This is a considerable amount of power, especially when we think of the availability of this power throughout the year. If the length of the coast line is, say 1200 Km, the power available is around 84 GW.

7.12 Equations for solar angles, along with a few simple measurements and an accurate clock, can be used for rough navigation purposes. Suppose you know it is June 22 ($n = 172$, the solstice), your watch, which is set for Pacific Standard Time tells you that (geometric) sunrise occurred at 4:23 A.M. and sunset was 15 hs 42 min later. Ignoring refraction and assuming you measured geometric sunrise and sunset (mid-point in the sun), do the following calculations to find your latitude and longitude.

- **a.** Knowing that solar noon occurs midway between sunrise and sunset, at what time would your watch tell you it is solar noon?
- **b.** Use (7.14) to determine your longitude.
- **c.** Use (7.17) to determine your latitude.

7.13 Following the procedure outlined in Example 7.7, you are to determine your location if on January 1, the newspaper says sunrise is 7:50 A.M. and sunset is at 3:50 P.M. (both Central Standard Time). Solar declination $\delta = -23.0^\circ$ and Figure 7.16 (or by calculation) $E = -3.6$ minutes.

- **a.** What clock time is solar noon?
- **b.** Use (7.14) to determine your longitude.
- **c.** Using (7.16), estimate the latitude without using the Q correction
- **d.** Estimate Q and from that find the clock time at which geometric sunrise occurs.
- **e.** Use (7.17) to determine your latitude.

7.14 A south-facing collector at latitude 40° is tipped up at an angle equal to its latitude. Compute the following insolations for January 1st at solar noon:

- **a.** The direct beam insolation normal to the sun's rays.
- **b.** Beam insolation on the collector.
- **c.** Diffuse radiation on the collector.
- **d.** Reflected radiation on the collector with ground reflectivity 0.2.

7.15 Create a "Clear Sky Insolation Calculator" for direct and diffuse radiation using the following spreadsheet as a guide. In this example, the insolation has been computed to be 964 W/m^2 for a South-facing collector tipped up at 45° at noon on November 7 at latitude 37.5°. Note the third column simply adjusts angles measured in degrees to radians.

Use the calculator to compute clear sky insolation under the following conditions:

- **a.** January 1, latitude 40°, horizontal insolation, solar noon
- **b.** March 21, latitude 20°, South-facing collector with tilt 20°, 11:00 A.M. (solar time)
- **c.** July 1, latitude 48°, South-East facing collector (azimuth 45°), tilt 20°, 2 P.M. (solar time)

Clear Sky Insolation Calculator:			
		radians	Radians = 0.017453292| × degrees
Day number n	311		ENTER (excel uses radians)
Latitude (L)	37.5	0.6545	ENTER
Collector azimuth (ϕ_C)	0	0	ENTER (+ is east of south)
Collector tilt (Σ)	45	0.7854	ENTER
Solar time (ST) (24 h)	12		ENTER
Hour angle H	0	0	$H = 15^\circ/\text{h} \times (12 - \text{ST})$ (7.10)
Declination (δ)	−17.11	−0.2986	$\delta = 23.45\sin(360/365\ (n - 81))$ (7.6)
Altitude angle (β)	35.39	0.6177	$\beta = \text{ASIN}((\cos L \cos\delta \cos H + \sin L \sin\delta)$ (7.8)
Solar azimuth (ϕ_S)	0.000	0	$\phi_S = \text{ASIN}(\cos\delta \sin H / \cos\beta)$ (7.9)
Air mass ratio (m)	1.727		$\text{m} = 1/\sin\beta$ (7.4)
A (W/m^2)	1204		$\text{A} = 1160 + 75\sin(360/365\ (n - 275))$ (7.22)
k	0.158		$k = 0.174 + 0.035\sin(360/365(n - 100))$ (7.23)
I_B (W/m^2)	917		$I_B = A\exp(-km)$ (7.21)
cos θ	0.986		$\cos\theta = \cos\beta\cos(\phi_s - \phi_c)\sin(\Sigma) + \sin\beta\cos(\Sigma)$ (7.26)
I_{BC} (W/m^2)	904		$I_{BC} = I_B\cos\theta$ (7.24)
C	0.076		$C = 0.095 + 0.04\ \sin(360/365(n - 100))$ (7.28)
I_{DC} (W/m^2)	60		$I_{DC} = C\,I_B(1 + \cos\Sigma)/2$ (7.29)
$I_C = I_{BC} + I_{DC}$ (W/m^2)	964		$I_C = I_{BC} + I_{DC}$ (ignoring reflection) (7.32)

7.16 Air Mass AM1.5 is supposedly the basis for a standard 1-sun insolation of 1 kW/m^2. To see whether this is reasonable, compute the following for a clear day on March 21st:

a. What solar altitude angle gives AM1.5?

b. What would be the direct beam radiation normal to the sun's rays?

c. What would be the diffuse radiation on a collector normal to the rays?

d. What would be the reflected radiation on a collector normal to the rays with $\rho = 0.2$?

e. What would be the total insolation normal to the rays?

7.17 Consider a comparison between a south-facing photovoltaic (PV) array with a tilt equal to its latitude located in Los Angeles versus one with a polar-mount, single-axis tracker. Assuming the PVs are 10% efficient at converting sunlight into electricity:

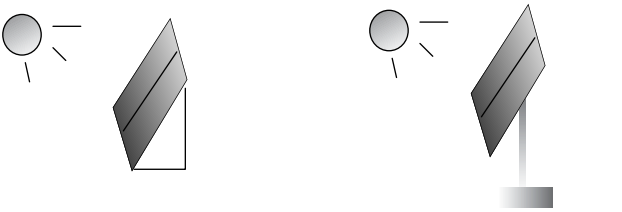


Figure P7.17

a. For a house that needs 4000-kWh per year, how large would each array need to be?

b. If the PVs cost \$400/m^2 and everything else in the two systems has the same cost except for the extra cost of the tracker, how much can the tracker cost (\$) to make the systems cost the same amount? How much per unit area of tracker (\$/m^2)?

c. Derive a general expression for the justifiable extra cost of a tracker per unit area (\$/m^2) as a function of the PV cost (\$/m^2) and the ratio of tracker insolation I_T to fixed insolation I_F.

CHAPTER 8

PHOTOVOLTAIC MATERIALS AND ELECTRICAL CHARACTERISTICS

8.1 INTRODUCTION

A material or device that is capable of converting the energy contained in photons of light into an electrical voltage and current is said to be *photovoltaic*. A photon with short enough wavelength and high enough energy can cause an electron in a photovoltaic material to break free of the atom that holds it. If a nearby electric field is provided, those electrons can be swept toward a metallic contact where they can emerge as an electric current. The driving force to power photovoltaics comes from the sun, and it is interesting to note that the surface of the earth receives something like 6000 times as much solar energy as our total energy demand.

The history of photovoltaics (PVs) began in 1839 when a 19-year-old French physicist, Edmund Becquerel, was able to cause a voltage to appear when he illuminated a metal electrode in a weak electrolyte solution (Becquerel, 1839). Almost 40 years later, Adams and Day were the first to study the photovoltaic effect in solids (Adams and Day, 1876). They were able to build cells made of selenium that were 1% to 2% efficient. Selenium cells were quickly adopted by the emerging photography industry for photometric light meters; in fact, they are still used for that purpose today.

As part of his development of quantum theory, Albert Einstein published a theoretical explanation of the photovoltaic effect in 1904, which led to a Nobel

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Prize in 1923. About the same time, in what would turn out to be a cornerstone of modern electronics in general, and photovoltaics in particular, a Polish scientist by the name of Czochralski began to develop a method to grow perfect crystals of silicon. By the 1940s and 1950s, the Czochralski process began to be used to make the first generation of single-crystal silicon photovoltaics, and that technique continues to dominate the photovoltaic (PV) industry today.

In the 1950s there were several attempts to commercialize PVs, but their cost was prohibitive. The real emergence of PVs as a practical energy source came in 1958 when they were first used in space for the Vanguard I satellite. For space vehicles, cost is much less important than weight and reliability, and solar cells have ever since played an important role in providing onboard power for satellites and other space craft. Spurred on by the emerging energy crises of the 1970s, the development work supported by the space program began to pay off back on the ground. By the late 1980s, higher efficiencies (Fig. 8.1) and lower costs (Fig. 8.2) brought PVs closer to reality, and they began to find application in many off-grid terrestrial applications such as pocket calculators, off-shore buoys, highway lights, signs and emergency call boxes, rural water pumping, and small home systems. While the amortized cost of photovoltaic power did drop dramatically in the 1990s, a decade later it is still about double what it needs to be to compete without subsidies in more general situations.

By 2002, worldwide production of photovoltaics had approached 600 MW per year and was increasing by over 40% per year (by comparison, global wind power sales were 10 times greater). However, as Fig. 8.3 shows, the U.S. share of this rapidly growing PV market has been declining and was, at the turn of the century,

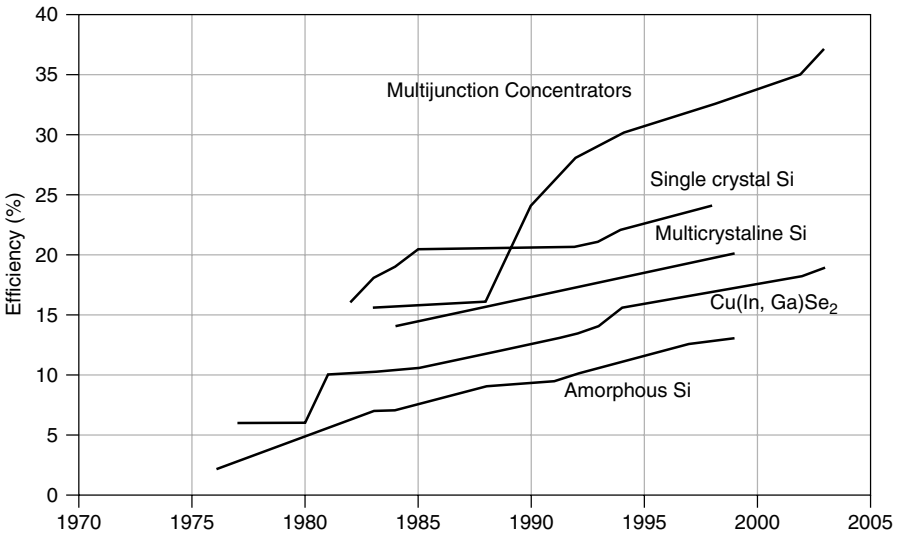


Figure 8.1 Best laboratory PV cell efficiencies for various technologies. (From National Center for Photovoltaics, www.nrel.gov/ncpv 2003).

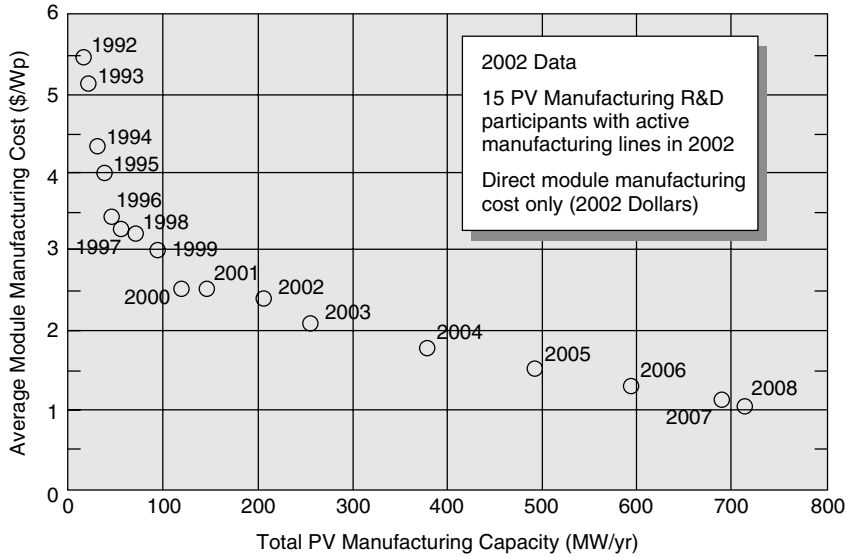


Figure 8.2 PV module manufacturing costs for DOE/US Industry Partners. Historical data through 2002, projections thereafter (www.nrel.gov/pvmat).

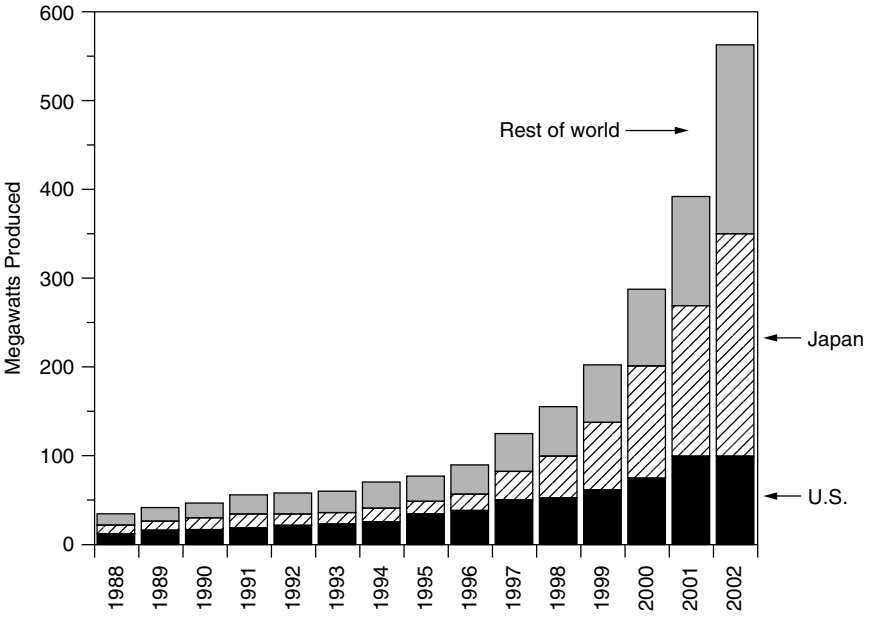


Figure 8.3 World production of photovoltaics is growing rapidly, but the U.S. share of the market is decreasing. Based on data from Maycock (2004).

less than 20% of the total. Critics of this decline point to the government's lack of enthusiasm to fund PV R&D. By comparison, Japan's R&D budget is almost an order of magnitude greater.

8.2 BASIC SEMICONDUCTOR PHYSICS

Photovoltaics use semiconductor materials to convert sunlight into electricity. The technology for doing so is very closely related to the solid-state technologies used to make transistors, diodes, and all of the other semiconductor devices that we use so many of these days. The starting point for most of the world's current generation of photovoltaic devices, as well as almost all semiconductors, is pure crystalline silicon. It is in the fourth column of the periodic table, which is referred to as Group IV (Table 8.1). Germanium is another Group IV element, and it too is used as a semiconductor in some electronics. Other elements that play important roles in photovoltaics are boldfaced. As we will see, boron and phosphorus, from Groups III and V, are added to silicon to make most PVs. Gallium and arsenic are used in GaAs solar cells, while cadmium and tellurium are used in CdTe cells.

Silicon has 14 protons in its nucleus, and so it has 14 orbital electrons as well. As shown in Fig. 8.4a, its outer orbit contains four valence electrons—that is, it is tetravalent. Those valence electrons are the only ones that matter in electronics, so it is common to draw silicon as if it has a +4 charge on its nucleus and four tightly held valence electrons, as shown in Fig. 8.4b.

In pure crystalline silicon, each atom forms covalent bonds with four adjacent atoms in the three-dimensional tetrahedral pattern shown in Fig. 8.5a. For convenience, that pattern is drawn as if it were all in a plane, as in Fig. 8.5b.

8.2.1 The Band Gap Energy

At absolute zero temperature, silicon is a perfect electrical insulator. There are no electrons free to roam around as there are in metals. As the temperature increases,

TABLE 8.1 The Portion of the Periodic Table of Greatest Importance for Photovoltaics Includes the Elements Silicon, Boron, Phosphorus, Gallium, Arsenic, Cadmium, and Tellurium

I	II	III	IV	V	VI
		5 B	6 C	7 N	8 O
		13 Al	**14 Si**	**15 P**	16 S
29 Cu	30 Zn	**31 Ga**	32 Ge	**33 As**	34 Se
47 Ag	**48 Cd**	49 In	50 Sn	51 Sb	**52 Te**

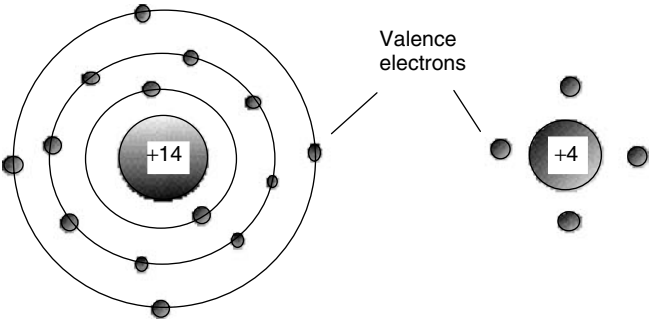


Figure 8.4 Silicon has 14 protons and electrons as in (a). A convenient shorthand is drawn in (b), in which only the four outer electrons are shown, spinning around a nucleus with a +4 charge.

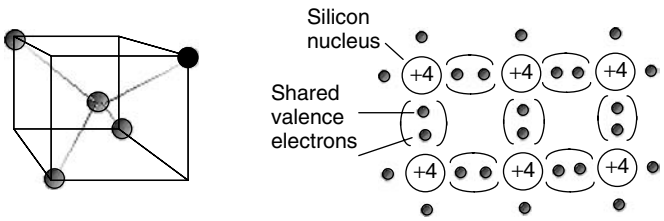


Figure 8.5 Crystalline silicon forms a three-dimensional tetrahedral structure (a); but it is easier to draw it as a two-dimensional flat array (b).

some electrons will be given enough energy to free themselves from their nuclei, making them available to flow as electric current. The warmer it gets, the more electrons there are to carry current, so its conductivity increases with temperature (in contrast to metals, where conductivity decreases). That change in conductivity, it turns out, can be used to advantage to make very accurate temperature sensors called *thermistors*. Silicon's conductivity at normal temperatures is still very low, and so it is referred to as a semiconductor. As we will see, by adding minute quantities of other materials, the conductivity of pure (intrinsic) semiconductors can be greatly increased.

Quantum theory describes the differences between conductors (metals) and semiconductors (e.g., silicon) using energy-band diagrams such as those shown in Fig. 8.6. Electrons have energies that must fit within certain allowable energy bands. The top energy band is called the conduction band, and it is electrons within this region that contribute to current flow. As shown in Fig. 8.6, the conduction band for metals is partially filled, but for semiconductors at absolute zero temperature, the conduction band is empty. At room temperature, only about one out of 10^{10} electrons in silicon exists in the conduction band.

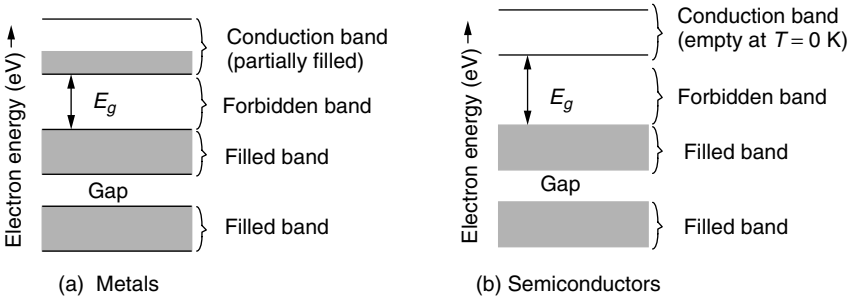


Figure 8.6 Energy bands for (a) metals and (b) semiconductors. Metals have partially filled conduction bands, which allows them to carry electric current easily. Semiconductors at absolute zero temperature have no electrons in the conduction band, which makes them insulators.

The gaps between allowable energy bands are called forbidden bands, the most important of which is the gap separating the conduction band from the highest filled band below it. The energy that an electron must acquire to jump across the forbidden band to the conduction band is called the band-gap energy, designated E_g. The units for band-gap energy are usually electron-volts (eV), where one electron-volt is the energy that an electron acquires when its voltage is increased by 1 V ($1\ \text{eV} = 1.6 \times 10^{-19}$ J).

The band-gap E_g for silicon is 1.12 eV, which means an electron needs to acquire that much energy to free itself from the electrostatic force that ties it to its own nucleus—that is, to jump into the conduction band. Where might that energy come from? We already know that a small number of electrons get that energy thermally. For photovoltaics, the energy source is photons of electromagnetic energy from the sun. When a photon with more than 1.12 eV of energy is absorbed by a solar cell, a single electron may jump to the conduction band. When it does so, it leaves behind a nucleus with a +4 charge that now has only three electrons attached to it. That is, there is a net positive charge, called a *hole*, associated with that nucleus as shown in Fig. 8.7a. Unless there is some way to sweep the electrons away from the holes, they will eventually recombine, obliterating both the hole and electron as in Fig. 8.7b. When recombination occurs, the energy that had been associated with the electron in the conduction band is released as a photon, which is the basis for light-emitting diodes (LEDs).

It is important to note that not only is the negatively charged electron in the conduction band free to roam around in the crystal, but the positively charged hole left behind can also move as well. A valence electron in a filled energy band can easily move to fill a hole in a nearby atom, without having to change energy bands. Having done so, the hole, in essence, moves to the nucleus from which the electron originated, as shown in Fig. 8.8. This is analogous to a student leaving her seat to get a drink of water. A roaming student (electron) and a seat (hole) are created. Another student already seated might decide he wants that newly

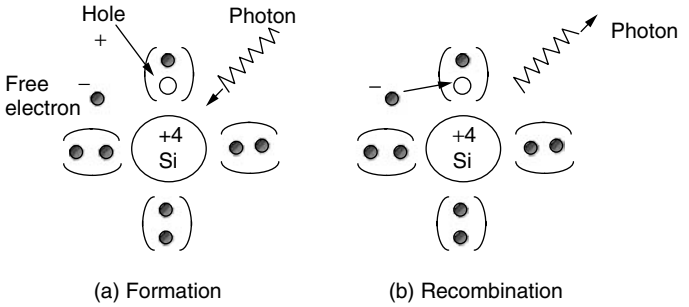


Figure 8.7 A photon with sufficient energy can create a hole–electron pair as in (a). The electron can recombine with the hole, releasing a photon of energy (b).

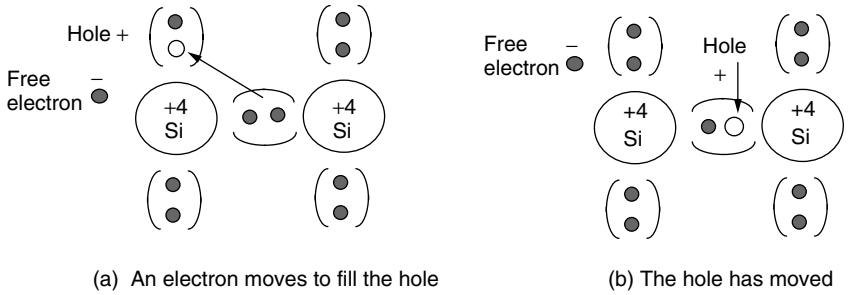


Figure 8.8 When a hole is filled by a nearby valence electron, the hole appears to move.

vacated seat, so he gets up and moves, leaving his seat behind. The empty seat appears to move around just the way a hole moves around in a semiconductor. The important point here is that electric current in a semiconductor can be carried not only by negatively charged electrons moving around, but also by positively charged holes that move around as well.

Thus, photons with enough energy create hole–electron pairs in a semiconductor. Photons can be characterized by their wavelengths or their frequency as well as by their energy; the three are related by the following:

$$c = \lambda \nu \tag{8.1}$$

where c is the speed of light (3×10^8 m/s), ν is the frequency (hertz), λ is the wavelength (m), and

$$E = h\nu = \frac{hc}{\lambda} \tag{8.2}$$

where E is the energy of a photon (J) and h is Planck's constant (6.626×10^{-34} J-s).

Example 8.1 Photons to Create Hole–Electron Pairs in Silicon What maximum wavelength can a photon have to create hole–electron pairs in silicon? What minimum frequency is that? Silicon has a band gap of 1.12 eV and 1 eV = 1.6×10^{-19} J.

Solution. From (8.2) the wavelength must be less than

$$\lambda \leq \frac{hc}{E} = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s} \times 3 \times 10^{8}\ \text{m/s}}{1.12\ \text{eV} \times 1.6 \times 10^{-19}\text{J/eV}} = 1.11 \times 10^{-6}\ \text{m} = 1.11\ \mu\text{m}$$

and from (8.1) the frequency must be at least

$$\nu \geq \frac{c}{\lambda} = \frac{3 \times 10^{8}\ \text{m/s}}{1.11 \times 10^{-6}\ \text{m}} = 2.7 \times 10^{14}\ \text{Hz}$$

For a silicon photovoltaic cell, photons with wavelength greater than 1.11 μm have energy $h\nu$ less than the 1.12-eV band-gap energy needed to excite an electron. None of those photons create hole–electron pairs capable of carrying current, so all of their energy is wasted. It just heats the cell. On the other hand, photons with wavelengths shorter than 1.11 μm have more than enough energy to excite an electron. Since one photon can excite *only* one electron, any extra energy above the 1.12 eV needed is also dissipated as waste heat in the cell. Figure 8.9 uses a plot of (8.2) to illustrate this important concept. The band gaps for other photovoltaic materials—gallium arsenide (GaAs), cadmium telluride (CdTe), and indium phosphide (InP), in addition to silicon—are shown in Table 8.2.

These two phenomena relating to photons with energies above and below the actual band gap establish a maximum theoretical efficiency for a solar cell. To explore this constraint, we need to introduce the solar spectrum.

8.2.2 The Solar Spectrum

As was described in the last chapter, the surface of the sun emits radiant energy with spectral characteristics that well match those of a 5800 K blackbody. Just outside of the earth's atmosphere, the average radiant flux is about 1.377 kW/m^2, an amount known as the solar constant. As solar radiation passes through the atmosphere, some is absorbed by various constituents in the atmosphere, so that by the time it reaches the earth's surface the spectrum is significantly distorted.

The amount of solar energy reaching the ground, as well as its spectral distribution, depends very much on how much atmosphere it has had to pass through to get there. Recall that the length of the path taken by the sun's rays through the atmosphere to reach a spot on the ground, divided by the path length corresponding to the sun directly overhead, is called the *air mass ratio*, m. Thus, an

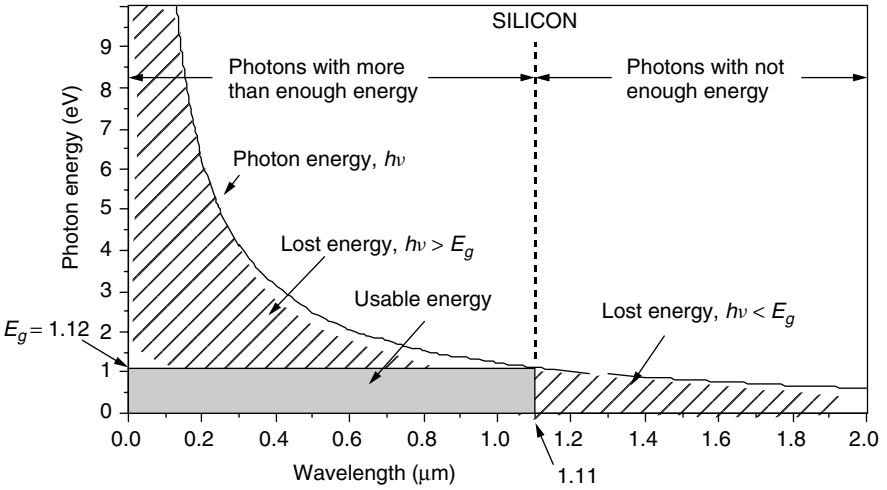


Figure 8.9 Photons with wavelengths above 1.11 μm don't have the 1.12 eV needed to excite an electron, and this energy is lost. Photons with shorter wavelengths have more than enough energy, but any energy above 1.12 eV is wasted as well.

TABLE 8.2 Band Gap and Cut-off Wavelength Above Which Electron Excitation Doesn't Occur

Quantity	Si	GaAs	CdTe	InP
Band gap (eV)	1.12	1.42	1.5	1.35
Cut-off wavelength (μm)	1.11	0.87	0.83	0.92

air mass ratio of 1 (designated "AM1") means that the sun is directly overhead. By convention, AM0 means no atmosphere; that is, it is the extraterrestrial solar spectrum. For most photovoltaic work, an air mass ratio of 1.5, corresponding to the sun being 42 degrees above the horizon, is assumed to be the standard. The solar spectrum at AM 1.5 is shown in Fig. 8.10. For an AM 1.5 spectrum, 2% of the incoming solar energy is in the UV portion of the spectrum, 54% is in the visible, and 44% is in the infrared.

8.2.3 Band-Gap Impact on Photovoltaic Efficiency

We can now make a simple estimate of the upper bound on the efficiency of a silicon solar cell. We know the band gap for silicon is 1.12 eV, corresponding to a wavelength of 1.11 μm, which means that any energy in the solar spectrum with wavelengths longer than 1.11 μm cannot send an electron into the conduction band. And, any photons with wavelength less than 1.11 μm waste their extra energy. If we know the solar spectrum, we can calculate the energy loss due to

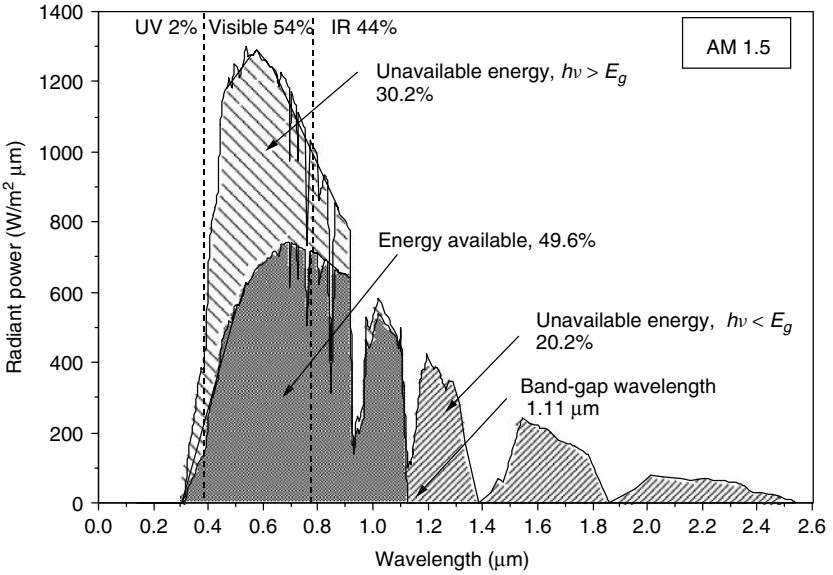


Figure 8.10 Solar spectrum at AM 1.5. Photons with wavelengths longer than 1.11 μm don't have enough energy to excite electrons (20.2% of the incoming solar energy); those with shorter wavelengths can't use all of their energy, which accounts for another 30.2% unavailable to a silicon photovoltaic cell. Spectrum is based on ERDA/NASA (1977).

these two fundamental constraints. Figure 8.10 shows the results of this analysis, assuming a standard air mass ratio AM 1.5. As is presented there, 20.2% of the energy in the spectrum is lost due to photons having less energy than the band gap of silicon ($h\nu < E_g$), and another 30.2% is lost due to photons with $h\nu > E_g$. The remaining 49.6% represents the maximum possible fraction of the sun's energy that could be collected with a silicon solar cell. That is, the constraints imposed by silicon's band gap limit the efficiency of silicon to just under 50%.

Even this simple discussion gives some insight into the trade-off between choosing a photovoltaic material that has a small band gap versus one with a large band gap. With a smaller band gap, more solar photons have the energy needed to excite electrons, which is good since it creates the charges that will enable current to flow. However, a small band gap means that more photons have surplus energy above the threshold needed to create hole–electron pairs, which wastes their potential. High band-gap materials have the opposite combination. A high band gap means that fewer photons have enough energy to create the current-carrying electrons and holes, which limits the current that can be generated. On the other hand, a high band gap gives those charges a higher voltage with less leftover surplus energy.

In other words, low band gap gives more current with less voltage while high band gap results in less current and higher voltage. Since power is the product

of current and voltage, there must be some middle-ground band gap, usually estimated to be between 1.2 eV and 1.8 eV, which will result in the highest power and efficiency. Figure 8.11 shows one estimate of the impact of band gap on the theoretical maximum efficiency of photovoltaics at both AM0 and AM1. The figure includes band gaps and maximum efficiencies for many of the most promising photovoltaic materials being developed today.

Notice that the efficiencies in Fig. 8.11 are roughly in the 20–25% range—well below the 49.6% we found when we considered only the losses caused by (a) photons with insufficient energy to push electrons into the conduction band and (b) photons with energy in excess of what is needed to do so. Other factors that contribute to the drop in theoretical efficiency include:

1. Only about half to two-thirds of the full band-gap voltage across the terminals of the solar cell.
2. Recombination of holes and electrons before they can contribute to current flow.
3. Photons that are not absorbed in the cell either because they are reflected off the face of the cell, or because they pass right through the cell, or because they are blocked by the metal conductors that collect current from the top of the cell.
4. Internal resistance within the cell, which dissipates power.

8.2.4 The *p*–*n* Junction

As long as a solar cell is exposed to photons with energies above the band-gap energy, hole–electron pairs will be created. The problem is, of course, that

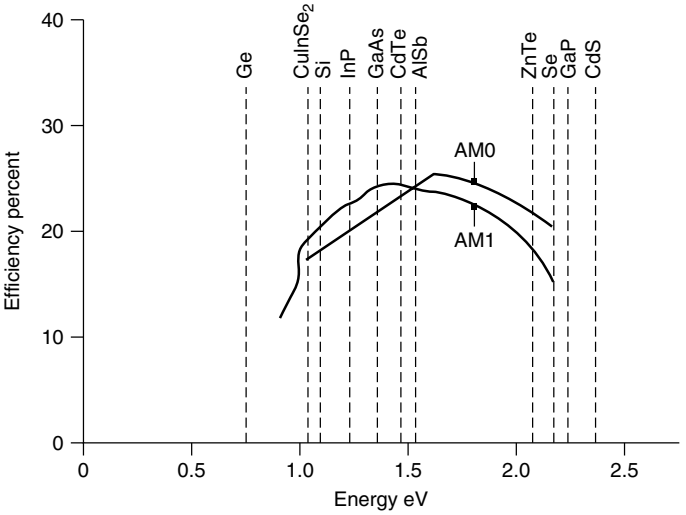


Figure 8.11 Maximum efficiency of photovoltaics as a function of their band gap. From Hersel and Zweibel (1982).

those electrons can fall right back into a hole, causing both charge carriers to disappear. To avoid that recombination, electrons in the conduction band must continuously be swept away from holes. In PVs this is accomplished by creating a built-in electric field within the semiconductor itself that pushes electrons in one direction and holes in the other. To create the electric field, two regions are established within the crystal. On one side of the dividing line separating the regions, pure (intrinsic) silicon is purposely contaminated with very small amounts of a trivalent element from column III of the periodic chart; on the other side, pentavalent atoms from column V are added.

Consider the side of the semiconductor that has been doped with a pentavalent element such as phosphorus. Only about 1 phosphorus atom per 1000 silicon atoms is typical. As shown in Fig. 8.12, an atom of the pentavalent impurity forms covalent bonds with four adjacent silicon atoms. Four of its five electrons are now tightly bound, but the fifth electron is left on its own to roam around the crystal. When that electron leaves the vicinity of its donor atom, there will remain a +5 donor ion fixed in the matrix, surrounded by only four negative valence electrons. That is, each donor atom can be represented as a single, fixed, immobile positive charge plus a freely roaming negative charge as shown in Fig. 8.12b. Pentavalent i.e., +5 elements donate electrons to their side of the semiconductor so they are called *donor* atoms. Since there are now negative charges that can move around the crystal, a semiconductor doped with donor atoms is referred to as an "*n-type material*."

On the other side of the semiconductor, silicon is doped with a trivalent element such as boron. Again the concentration of dopants is small, something on the order of 1 boron atom per 10 million silicon atoms. These dopant atoms fall into place in the crystal, forming covalent bonds with the adjacent silicon atoms as shown in Fig. 8.13. Since each of these impurity atoms has only three electrons, only three of the covalent bonds are filled, which means that a positively charged hole appears next to its nucleus. An electron from a neighboring silicon atom can easily move into the hole, so these impurities are referred to as *acceptors* since they accept electrons. The filled hole now means there are four negative charges

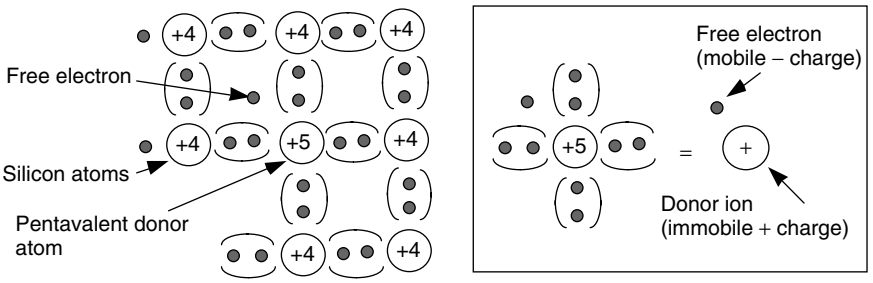
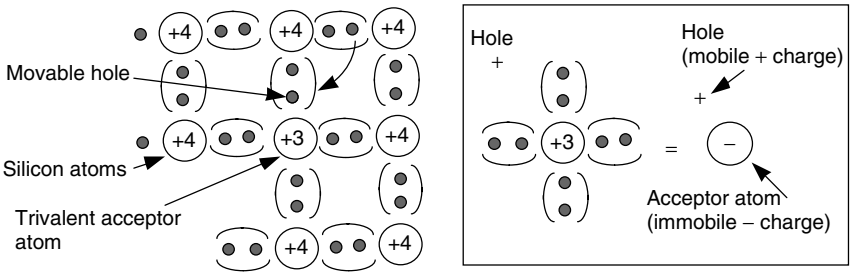
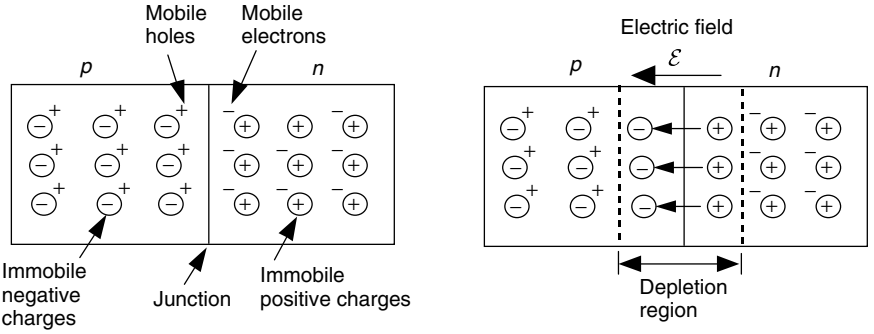


Figure 8.12 An *n*-type material. (a) The pentavalent donor. (b) The representation of the donor as a mobile negative charge with a fixed, immobile positive charge.



(a) An acceptor atom in Si crystal (b) Representation of the acceptor atom

Figure 8.13 In a p-type material, trivalent acceptors contribute movable, positively charged holes leaving rigid, immobile negative charges in the crystal lattice.



(a) When first brought together (b) In steady-state

Figure 8.14 (a) When a p–n junction is first formed, there are mobile holes in the p-side and mobile electrons in the n-side. (b) As they migrate across the junction, an electric field builds up that opposes, and quickly stops, diffusion.

surrounding a +3 nucleus. All four covalent bonds are now filled creating a fixed, immobile net negative charge at each acceptor atom. Meanwhile, each acceptor has created a positively charged hole that is free to move around in the crystal, so this side of the semiconductor is called a *p-type* material.

Now, suppose we put an n-type material next to a p-type material forming a junction between them. In the n-type material, mobile electrons drift by diffusion across the junction. In the p-type material, mobile holes drift by diffusion across the junction in the opposite direction. As depicted in Fig. 8.14, when an electron crosses the junction it fills a hole, leaving an immobile, positive charge behind in the n-region, while it creates an immobile, negative charge in the p-region. These immobile charged atoms in the p and n regions create an electric field that works against the continued movement of electrons and holes across the junction. As the diffusion process continues, the electric field countering that movement increases until eventually (actually, almost instantaneously) all further movement of charged carriers across the junction stops.

The exposed immobile charges creating the electric field in the vicinity of the junction form what is called a *depletion region*, meaning that the mobile charges are depleted—gone—from this region. The width of the depletion region is only about 1 μm and the voltage across it is perhaps 1 V, which means the field strength is about 10,000 V/cm! Following convention, the arrows representing an electric field in Fig. 8.14b start on a positive charge and end on a negative charge. The arrow, therefore, points in the direction that the field would push a positive charge, which means that it holds the mobile positive holes in the p-region (while it repels the electrons back into the n-region).

8.2.5 The *p–n* Junction Diode

Anyone familiar with semiconductors will immediately recognize that what has been described thus far is just a common, conventional $p-n$ junction diode, the characteristics of which are presented in Fig. 8.15. If we were to apply a voltage V_d across the diode terminals, forward current would flow easily through the diode from the p-side to the n-side; but if we try to send current in the reverse direction, only a very small ($\approx 10^{-12}$ A/cm^2) reverse saturation current I_0 will flow. This reverse saturation current is the result of thermally generated carriers with the holes being swept into the p-side and the electrons into the n-side. In the forward direction, the voltage drop across the diode is only a few tenths of a volt.

The symbol for a real diode is shown here as a blackened triangle with a bar; the triangle suggests an arrow, which is a convenient reminder of the direction in which current flows easily. The triangle is blackened to distinguish it from an "ideal" diode. Ideal diodes have no voltage drop across them in the forward direction, and no current at all flows in the reverse direction.

The voltage–current characteristic curve for the $p-n$ junction diode is described by the following Shockley diode equation:

$$I_d = I_0(e^{qV_d/kT} - 1) \tag{8.3}$$

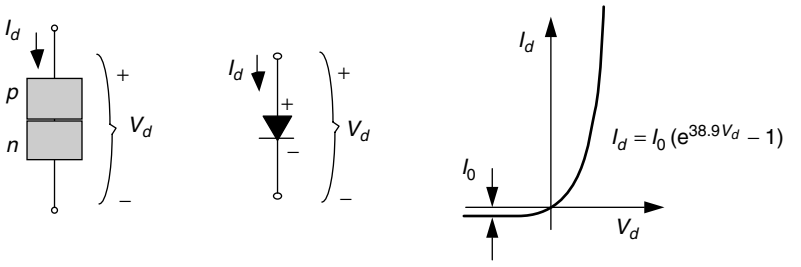


Figure 8.15 A $p-n$ junction diode allows current to flow easily from the p-side to the n-side, but not in reverse. (a) $p-n$ junction; (b) its symbol; (c) its characteristic curve.

where I_d is the diode current in the direction of the arrow (A), V_d is the voltage across the diode terminals from the p-side to the n-side (V), I_0 is the reverse saturation current (A), q is the electron charge (1.602×10^{-19}C), k is Boltzmann's constant (1.381×10^{-23} J/K), and T is the junction temperature (K).

Substituting the above constants into the exponent of (8.3) gives

$$\frac{qV_d}{kT} = \frac{1.602 \times 10^{-19}}{1.381 \times 10^{-23}} \cdot \frac{V_d}{T(\text{K})} = 11,600 \frac{V_d}{T(\text{K})} \tag{8.4}$$

A junction temperature of 25°C is often used as a standard, which results in the following diode equation:

$$I_d = I_0(e^{38.9V_d} - 1) \qquad (\text{at } 25^\circ\text{C}) \tag{8.5}$$

Example 8.2 A $p-n$ Junction Diode. Consider a $p-n$ junction diode at 25°C with a reverse saturation current of 10^{-9} A. Find the voltage drop across the diode when it is carrying the following:

a. no current (open-circuit voltage)
b. 1 A
c. 10 A

Solution

a. In the open-circuit condition, $I_d = 0$, so from (8.5) $V_d = 0$.
b. With $I_d = 1$ A, we can find V_d by rearranging (8.5):

$$V_d = \frac{1}{38.9} \ln\left(\frac{I_d}{I_0} + 1\right) = \frac{1}{38.9} \ln\left(\frac{1}{10^{-9}} + 1\right) = 0.532 \text{ V}$$

c. with $I_d = 10$ A,

$$V_d = \frac{1}{38.9} \ln\left(\frac{10}{10^{-9}} + 1\right) = 0.592 \text{ V}$$

Notice how little the voltage drop changes as the diode conducts more and more current, changing by only about 0.06 V as the current increased by a factor of 10. Often in normal electronic circuit analysis, the diode voltage drop when it is conducting current is assumed to be nominally about 0.6 V, which is quite in line with the above results.

While the Shockley diode equation (8.3) is appropriate for our purposes, it should be noted that in some circumstances it is modified with an "ideality

factor" A, which accounts for different mechanisms responsible for moving carriers across the junction. The resulting equation is then

$$I_d = I_0(e^{qV_d/AkT} - 1) \tag{8.6}$$

where the ideality factor A is 1 if the transport process is purely diffusion, and $A \approx 2$ if it is primarily recombination in the depletion region.

8.3 A GENERIC PHOTOVOLTAIC CELL

Let us consider what happens in the vicinity of a p–n junction when it is exposed to sunlight. As photons are absorbed, hole-electron pairs may be formed. If these mobile charge carriers reach the vicinity of the junction, the electric field in the depletion region will push the holes into the p-side and push the electrons into the n-side, as shown in Fig. 8.16. The p-side accumulates holes and the n-side accumulates electrons, which creates a voltage that can be used to deliver current to a load.

If electrical contacts are attached to the top and bottom of the cell, electrons will flow out of the n-side into the connecting wire, through the load and back to the p-side as shown in Fig. 8.17. Since wire cannot conduct holes, it is only the electrons that actually move around the circuit. When they reach the p-side, they recombine with holes completing the circuit. By convention, positive current flows in the direction opposite to electron flow, so the current arrow in the figure shows current going from the p-side to the load and back into the n-side.

8.3.1 The Simplest Equivalent Circuit for a Photovoltaic Cell

A simple equivalent circuit model for a photovoltaic cell consists of a real diode in parallel with an ideal current source as shown in Fig. 8.18. The ideal current source delivers current in proportion to the solar flux to which it is exposed.

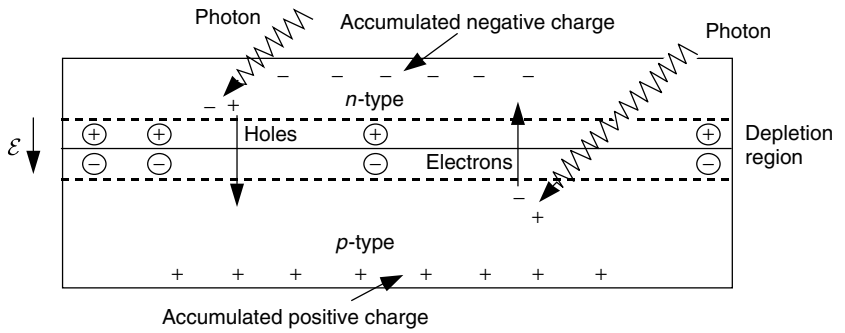


Figure 8.16 When photons create hole–electron pairs near the junction, the electric field in the depletion region sweeps holes into the p-side and sweeps electrons into the n-side of the cell.

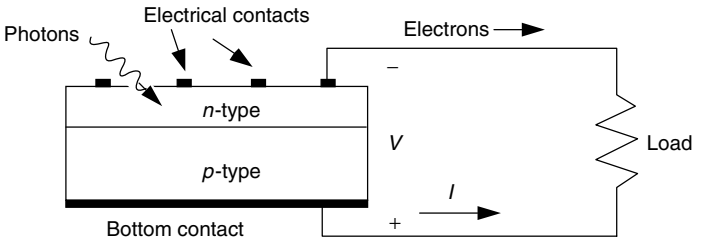


Figure 8.17 Electrons flow from the n-side contact, through the load, and back to the p-side where they recombine with holes. Conventional current I is in the opposite direction.

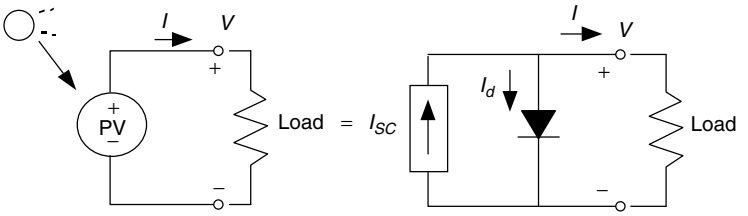


Figure 8.18 A simple equivalent circuit for a photovoltaic cell consists of a current source driven by sunlight in parallel with a real diode.

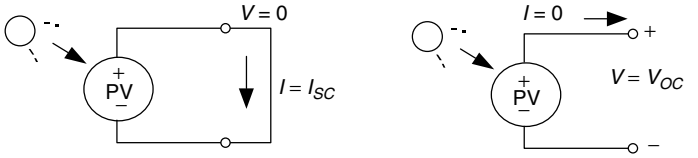


Figure 8.19 Two important parameters for photovoltaics are the short-circuit current I_{SC} and the open-circuit voltage V_{OC}.

There are two conditions of particular interest for the actual PV and for its equivalent circuit. As shown in Fig. 8.19, they are: (1) the current that flows when the terminals are shorted together (the short-circuit current, I_{SC}) and (2) the voltage across the terminals when the leads are left open (the open-circuit voltage, V_{OC}). When the leads of the equivalent circuit for the PV cell are shorted together, no current flows in the (real) diode since $V_d = 0$, so all of the current from the ideal source flows through the shorted leads. Since that short-circuit current must equal I_{SC}, the magnitude of the ideal current source itself must be equal to I_{SC}.

Now we can write a voltage and current equation for the equivalent circuit of the PV cell shown in Fig. 8.18b. Start with

$$I = I_{SC} - I_d \tag{8.7}$$

and then substitute (8.3) into (8.7) to get

$$I = I_{SC} - I_0 \left(e^{qV/kT} - 1\right) \tag{8.8}$$

It is interesting to note that the second term in (8.8) is just the diode equation with a negative sign. That means that a plot of (8.8) is just I_{SC} added to the diode curve of Fig. 8.15c turned upside-down. Figure 8.20 shows the current–voltage relationship for a PV cell when it is dark (no illumination) and light (illuminated) based on (8.8).

When the leads from the PV cell are left open, $I = 0$ and we can solve (8.8) for the open-circuit voltage V_{OC} :

$$V_{OC} = \frac{kT}{q} \ln\left(\frac{I_{SC}}{I_0} + 1\right) \tag{8.9}$$

And at 25°C, (8.8) and (8.9) become

$$I = I_{SC} - I_0(e^{38.9\ \mathrm{V}} - 1) \tag{8.10}$$

and

$$V_{OC} = 0.0257 \ln\left(\frac{I_{SC}}{I_0} + 1\right) \tag{8.11}$$

In both of these equations, short-circuit current, I_{SC}, is directly proportional to solar insolation, which means that we can now quite easily plot sets of PV current–voltage curves for varying sunlight. Also, quite often laboratory specifications for the performance of photovoltaics are given per cm^2 of junction

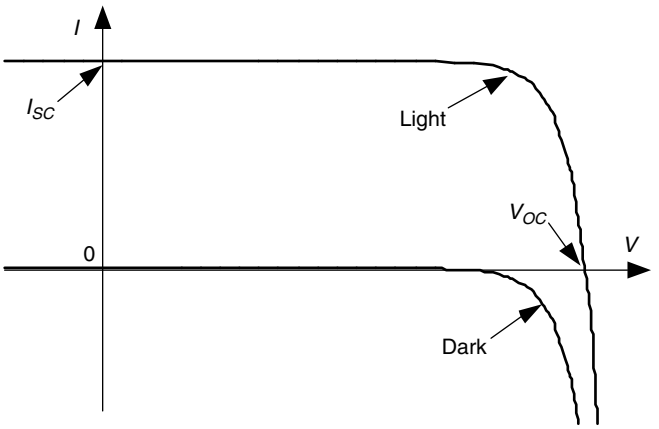


Figure 8.20 Photovoltaic current–voltage relationship for "dark" (no sunlight) and "light" (an illuminated cell). The dark curve is just the diode curve turned upside-down. The light curve is the dark curve plus I_{SC}.

area, in which case the currents in the above equations are written as current densities. Both of these points are illustrated in the following example.

Example 8.3 The $I-V$ Curve for a Photovoltaic Cell. Consider a 100-cm^2 photovoltaic cell with reverse saturation current $I_0 = 10^{-12}$ A/cm^2. In full sun, it produces a short-circuit current of 40 mA/cm^2 at 25°C. Find the open-circuit voltage at full sun and again for 50% sunlight. Plot the results.

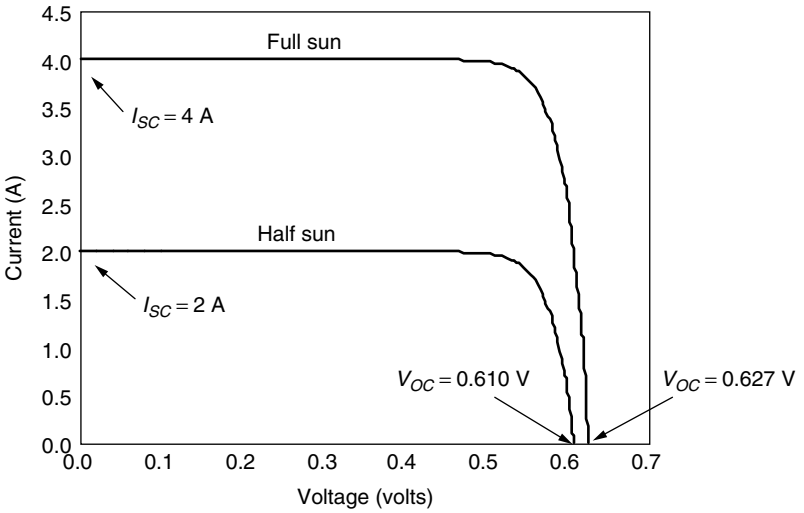
Solution. The reverse saturation current I_0 is 10^{-12} A/cm^2 $\times$ 100 cm^2 $= 1 \times 10^{-10}$ A. At full sun I_{SC} is 0.040 A/cm^2 $\times$ 100 cm^2 $= 4.0$ A. From (8.11) the open-circuit voltage is

$$V_{OC} = 0.0257 \ \ln\left(\frac{I_{SC}}{I_0} + 1\right) = 0.0257 \ln\left(\frac{4.0}{10^{-10}} + 1\right) = 0.627 \text{ V}$$

Since short-circuit current is proportional to solar intensity, at half sun $I_{SC} = 2$ A and the open-circuit voltage is

$$V_{OC} = 0.0257 \ln\left(\frac{2}{10^{-10}} + 1\right) = 0.610 \text{ V}$$

Plotting (8.10) gives us the following:



8.3.2 A More Accurate Equivalent Circuit for a PV Cell

There are times when a more complex PV equivalent circuit than the one shown in Fig. 8.18 is needed. For example, consider the impact of shading on a string of cells wired in series (Fig. 8.21 shows two such cells). If any cell in the string is in the dark (shaded), it produces no current. In our simplified equivalent circuit for the shaded cell, the current through that cell's current source is zero and its diode is back biased so it doesn't pass any current either (other than a tiny amount of reverse saturation current). This means that the simple equivalent circuit suggests that no power will be delivered to a load if any of its cells are shaded. While it is true that PV modules are very sensitive to shading, the situation is not quite as bad as that. So, we need a more complex model if we are going to be able to deal with realities such as the shading problem.

Figure 8.22 shows a PV equivalent circuit that includes some parallel leakage resistance R_p. The ideal current source I_{SC} in this case delivers current to the diode, the parallel resistance, and the load:

$$I = (I_{SC} - I_d) - \frac{V}{R_p} \tag{8.12}$$

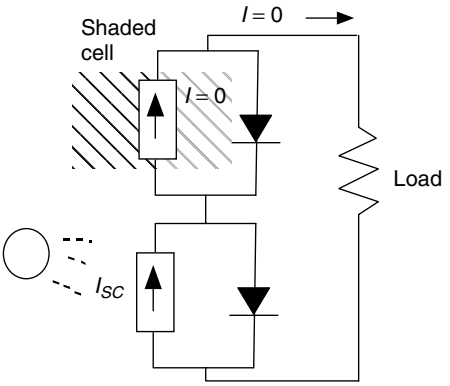


Figure 8.21 The simple equivalent circuit of a string of cells in series suggests no current can flow to the load if any cell is in the dark (shaded). A more complex model can deal with this problem.

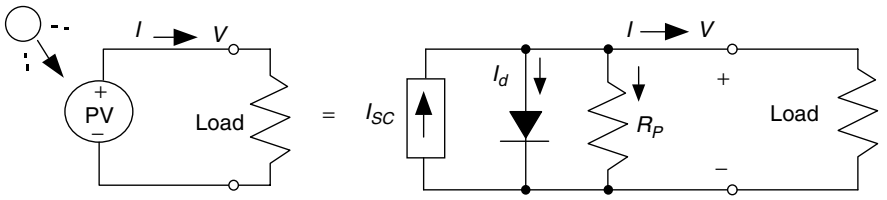


Figure 8.22 The simple PV equivalent circuit with an added parallel resistance.

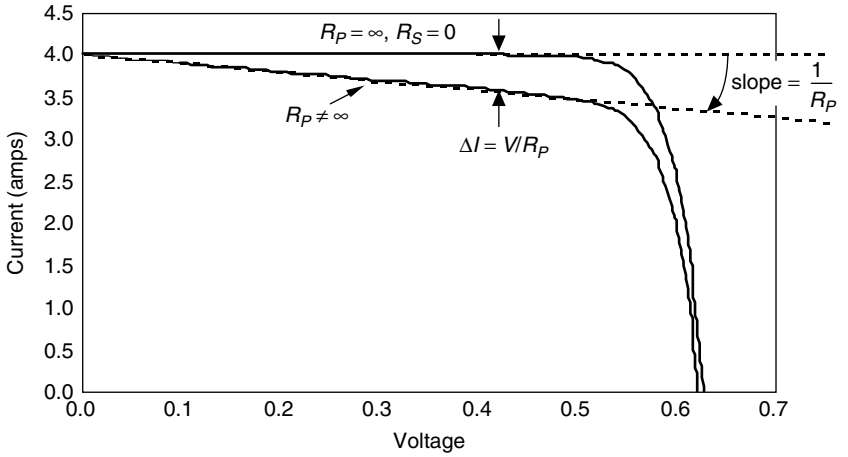


Figure 8.23 Modifying the idealized PV equivalent circuit by adding parallel resistance causes the current at any given voltage to drop by V/R_P.

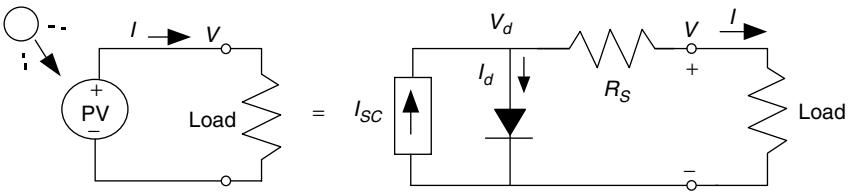


Figure 8.24 A PV equivalent circuit with series resistance.

The term in the parentheses of (8.12) is the same current that we had for the simple model. So, what (8.12) tells us is that at any given voltage, the parallel leakage resistance causes load current for the ideal model to be decreased by V/R_p as is shown in Fig. 8.23.

For a cell to have losses of less than 1% due to its parallel resistance, R_P should be greater than about

$$R_P > \frac{100V_{OC}}{I_{SC}} \tag{8.13}$$

For a large cell, I_{SC} might be around 7 A and V_{OC} might be about 0.6 V, which says its parallel resistance should be greater than about 9 Ω.

An even better equivalent circuit will include series resistance as well as parallel resistance. Before we can develop that model, consider Fig. 8.24 in which the original PV equivalent circuit has been modified to just include some series resistance, R_S. Some of this might be contact resistance associated with the bond between the cell and its wire leads, and some might be due to the resistance of the semiconductor itself.

To analyze Fig. 8.24, start with the simple equivalent circuit (8.8)

$$I = I_{SC} - I_d = I_{SC} - I_0\left(e^{qV_d/kT} - 1\right) \tag{8.8}$$

and then add the impact of R_S,

$$V_d = V + I \cdot R_S \tag{8.14}$$

to give

$$I = I_{SC} - I_0\left\{\exp\left[\frac{q(V + I \cdot R_S)}{kT}\right] - 1\right\} \tag{8.15}$$

Equation (8.15) can be interpreted as the original PV $I-V$ curve with the voltage at any given current shifted to the left by $\Delta V = IR_S$ as shown in Fig. 8.25.

For a cell to have less than 1% losses due to the series resistance, R_S will need to be less than about

$$R_S < \frac{0.01V_{OC}}{I_{SC}} \tag{8.16}$$

which, for a large cell with $I_{SC} = 7$ A and $V_{OC} = 0.6$ V, would be less than 0.0009 Ω.

Finally, let us generalize the PV equivalent circuit by including both series and parallel resistances as shown in Fig. 8.26. We can write the following equation for current and voltage:

$$I = I_{SC} - I_0\left\{\exp\left[\frac{q(V + I \cdot R_S)}{kT}\right] - 1\right\} - \left(\frac{V + I \cdot R_S}{R_P}\right) \tag{8.17}$$

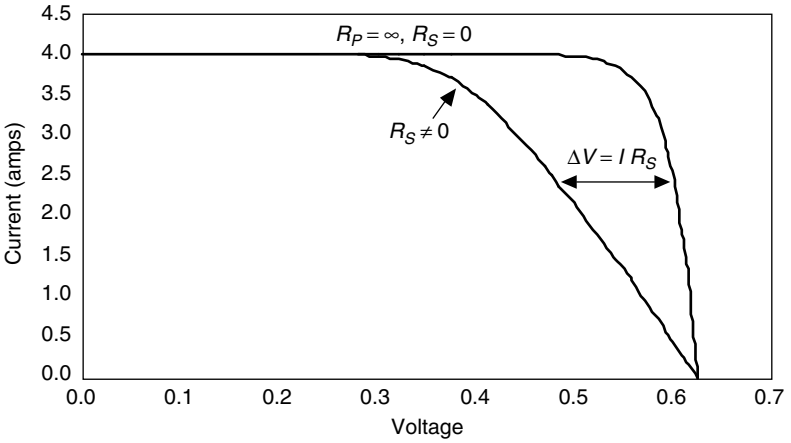


Figure 8.25 Adding series resistance to the PV equivalent circuit causes the voltage at any given current to shift to the left by $\Delta V = IR_S$.

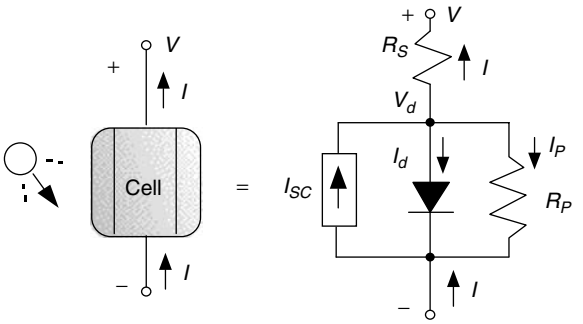


Figure 8.26 A more complex equivalent circuit for a PV cell includes both parallel and series resistances. The shaded diode reminds us that this is a "real" diode rather than an ideal one.

Under the standard assumption of a 25°C cell temperature, (8.17) becomes

$$I = I_{SC} - I_0\left[e^{38.9(V+IR_S)} - 1\right] - \frac{1}{R_P}(V + IR_S) \qquad \text{at } 25^\circ\text{C} \tag{8.18}$$

Unfortunately, (8.18) is a complex equation for which there is no explicit solution for either voltage V or current I. A spreadsheet solution, however, is fairly straightforward and has the extra advantage of enabling a graph of I versus V to be obtained easily. The approach is based on incrementing values of diode voltage, V_d, in the spreadsheet. For each value of V_d, corresponding values of current I and voltage V can easily be found.

Using the sign convention shown in Fig. 8.26 and applying Kirchhoff's Current Law to the node above the diode, we can write

$$I_{SC} = I + I_d + I_P \tag{8.19}$$

Rearranging, and substituting the Shockley diode equation (8.5) at 25°C gives

$$I = I_{SC} - \mathrm{I}_0(e^{38.9V_d} - 1) - \frac{V_d}{R_P} \tag{8.20}$$

With an assumed value of V_d in a spreadsheet, current I can be found from (8.20). Voltage across an individual cell then can be found from

$$V = V_d - IR_S \tag{8.21}$$

A plot of (8.18) obtained this way for an equivalent circuit with $R_S = 0.05\ \Omega$ and $R_P = 1\ \Omega$ is shown in Fig. 8.27. As might be expected, the graph combines features of Fig. 8.23 and 8.25.

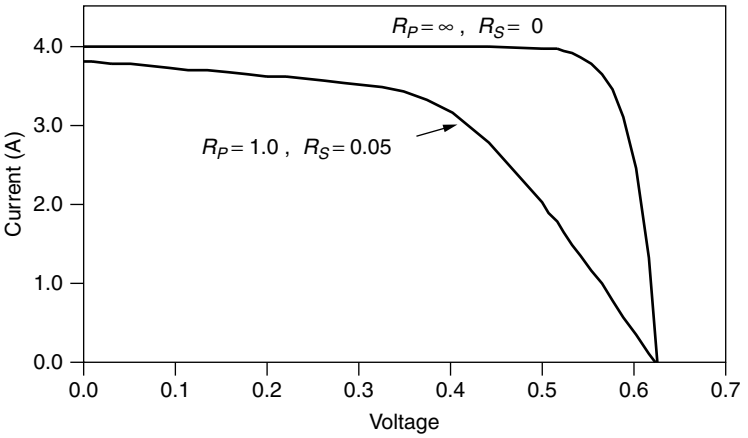


Figure 8.27 Series and parallel resistances in the PV equivalent circuit decrease both voltage and current delivered. To improve cell performance, high R_P and low R_S are needed.

8.4 FROM CELLS TO MODULES TO ARRAYS

Since an individual cell produces only about 0.5 V, it is a rare application for which just a single cell is of any use. Instead, the basic building block for PV applications is a *module* consisting of a number of pre-wired cells in series, all encased in tough, weather-resistant packages. A typical module has 36 cells in series and is often designated as a "12-V module" even though it is capable of delivering much higher voltages than that. Some 12-V modules have only 33 cells, which, as will be seen later may, be desirable in certain very simple battery charging systems. Large 72-cell modules are now quite common, some of which have all of the cells wired in series, in which case they are referred to as 24-V modules. Some 72-cell modules can be field-wired to act either as 24-V modules with all 72 cells in series or as 12-V modules with two parallel strings having 36 series cells in each.

Multiple modules, in turn, can be wired in series to increase voltage and in parallel to increase current, the product of which is power. An important element in PV system design is deciding how many modules should be connected in series and how many in parallel to deliver whatever energy is needed. Such combinations of modules are referred to as an *array*. Figure 8.28 shows this distinction between cells, modules, and arrays.

8.4.1 From Cells to a Module

When photovoltaics are wired in series, they all carry the same current, and at any given current their voltages add as shown in Fig. 8.29. That means we can continue the spreadsheet solution of (8.18) to find an overall module voltage

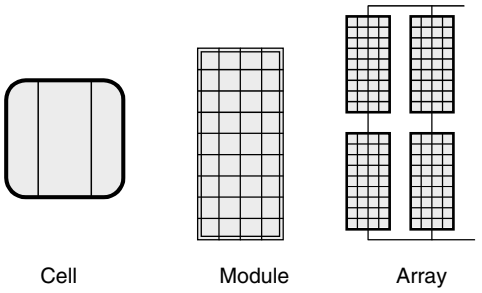


Figure 8.28 Photovoltaic cells, modules, and arrays.

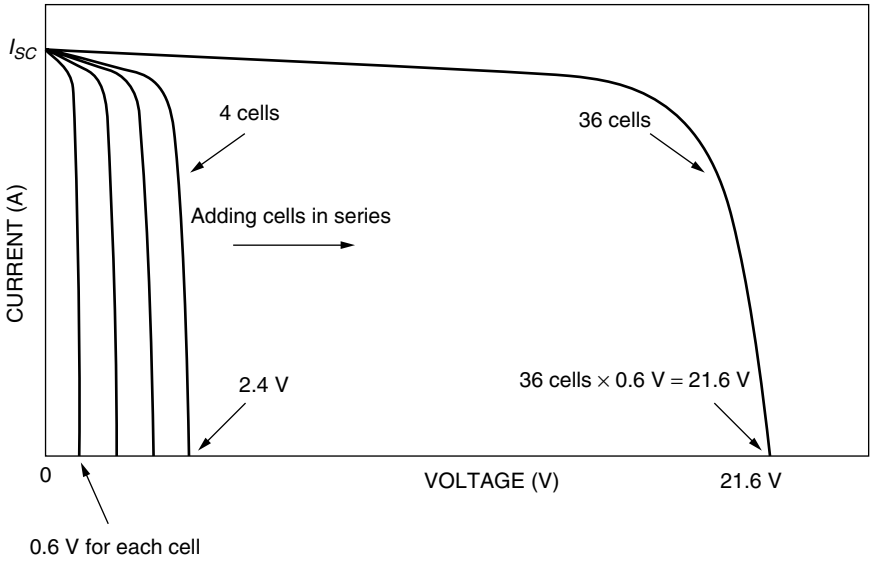


Figure 8.29 For cells wired in series, their voltages at any given current add. A typical module will have 36 cells.

V_{module} by multiplying (8.21) by the number of cells in the module n.

$$V_{\text{module}} = n(V_d - IR_S) \tag{8.22}$$

Example 8.4 Voltage and Current from a PV Module. A PV module is made up of 36 identical cells, all wired in series. With 1-sun insolation (1 kW/m^2), each cell has short-circuit current $I_{SC} = 3.4$ A and at 25°C its reverse saturation current is $I_0 = 6 \times 10^{-10}$ A. Parallel resistance $R_P = 6.6\ \Omega$ and series resistance $R_S = 0.005\ \Omega$.

a. Find the voltage, current, and power delivered when the junction voltage of each cell is 0.50 V.
b. Set up a spreadsheet for I and V and present a few lines of output to show how it works.

Solution.

a. Using $V_d = 0.50$ V in (8.20) along with the other data gives current:

$$I = I_{SC} - I_0(e^{38.9V_d} - 1) - \frac{V_d}{R_P}$$
$$= 3.4 - 6 \times 10^{-10}(e^{38.9\times 0.50} - 1) - \frac{0.50}{6.6} = 3.16 \text{ A}$$

Under these conditions, (8.22) gives the voltage produced by the 36-cell module:

$$V_{\text{module}} = n(V_d - IR_S) = 36(0.50 - 3.16 \times 0.005) = 17.43 \text{ V}$$

Power delivered is therefore

$$P(\text{watts}) = V_{\text{module}}\ I = 17.43 \times 3.16 = 55.0 \text{ W}$$

b. A spreadsheet might look something like the following:

Number of cells, $n = 36$
Parallel resistance/cell R_P (ohms) $= 6.6$
Series resistance/cell R_S (ohms) $= 0.005$
Reverse saturation current I_0 (A) $= 6.00\text{E-}10$
Short-circuit current at 1-sun (A) $= 3.4$

V_d	$I = I_{SC} - I_0\left(e^{38.9V_d} - 1\right) - \frac{V_d}{R_p}$	$V_{\text{module}} = n(V_d - IR_S)$	P (watts) $= V_{\text{module}}I$
0.49	3.21	17.06	54.80
0.50	3.16	17.43	55.02
0.51	3.07	17.81	54.75
0.52	2.96	18.19	53.76
0.53	2.78	18.58	51.65
0.54	2.52	18.99	47.89
0.55	2.14	19.41	41.59

Notice that we have found the maximum power point for this module, which is at $I = 3.16$ A, $V = 17.43$ V, and $P = 55$ W. This would be described as a 55-W module.

8.4.2 From Modules to Arrays

Modules can be wired in series to increase voltage, and in parallel to increase current. Arrays are made up of some combination of series and parallel modules to increase power.

For modules in series, the $I-V$ curves are simply added along the voltage axis. That is, at any given current (which flows through each of the modules), the total voltage is just the sum of the individual module voltages as is suggested in Fig. 8.30.

For modules in parallel, the same voltage is across each module and the total current is the sum of the currents. That is, at any given voltage, the $I-V$ curve of the parallel combination is just the sum of the individual module currents at that voltage. Figure 8.31 shows the $I-V$ curve for three modules in parallel.

When high power is needed, the array will usually consist of a combination of series and parallel modules for which the total $I-V$ curve is the sum of the individual module $I-V$ curves. There are two ways to imagine wiring a series/parallel combination of modules: The series modules may be wired as strings, and the strings wired in parallel as in Fig. 8.32a, or the parallel modules may be wired together first and those units combined in series as in 8.32b. The total $I-V$ curve is just the sum of the individual module curves, which is the same in either case when everything is working right. There is a reason, however, to prefer the wiring of strings in parallel (Fig. 8.32a). If an entire string is removed from service for some reason, the array can still deliver whatever voltage is needed by the load, though the current is diminished, which is not the case when a parallel group of modules is removed.

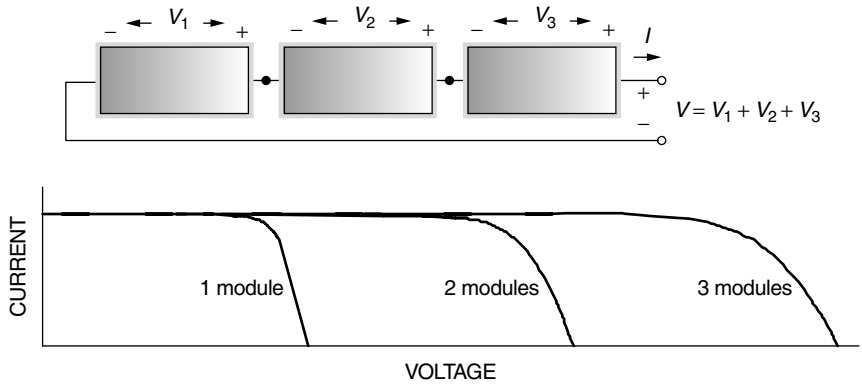


Figure 8.30 For modules in series, at any given current the voltages add.

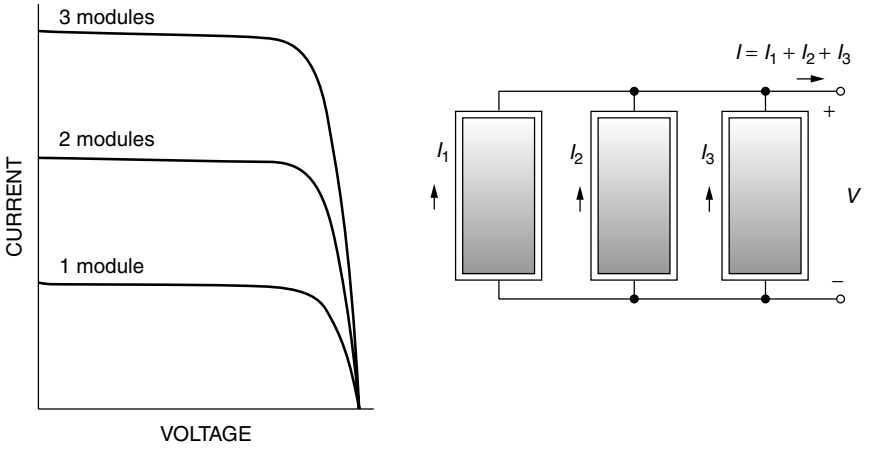


Figure 8.31 For modules in parallel, at any given voltage the currents add.

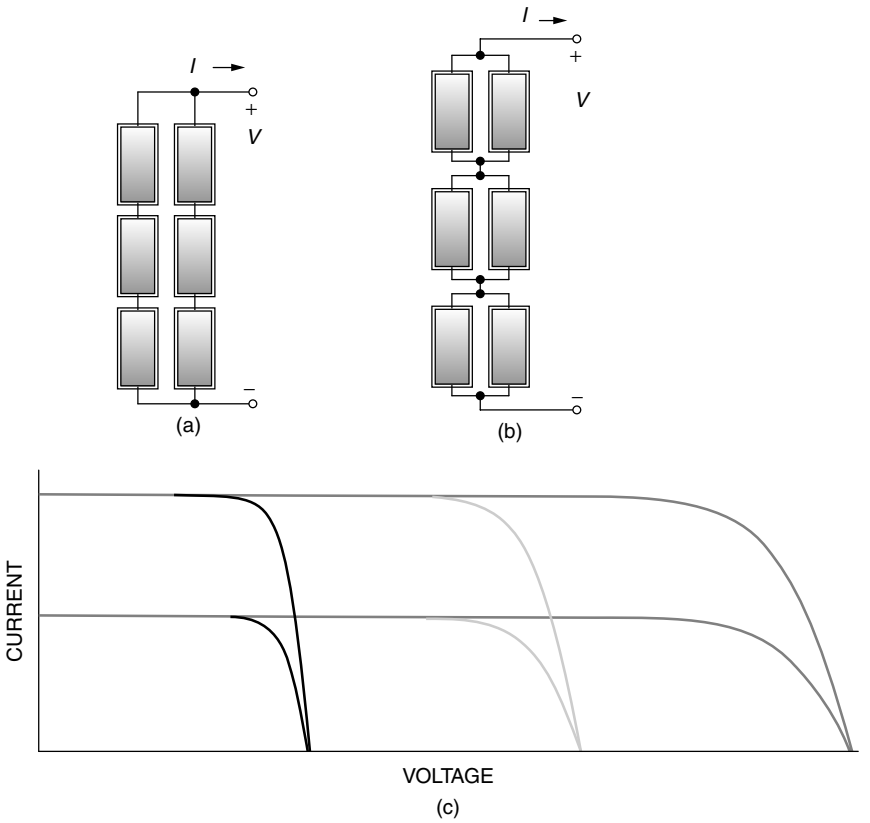


Figure 8.32 Two ways to wire an array with three modules in series and two modules in parallel. Although the $I-V$ curves for arrays are the same, two strings of three modules each (a) is preferred. The total $I-V$ curve of the array is shown in (c).

8.5 THE PV $I-V$ CURVE UNDER STANDARD TEST CONDITIONS (STC)

Consider, for the moment, a single PV module that you want to connect to some sort of a load (Fig. 8.33). The load might be a dc motor driving a pump or it might be a battery, for example. Before the load is connected, the module sitting in the sun will produce an open-circuit voltage V_{OC}, but no current will flow. If the terminals of the module are shorted together (which doesn't hurt the module at all, by the way), the short-circuit current I_{SC} will flow, but the output voltage will be zero. In both cases, since power is the product of current and voltage, no power is delivered by the module and no power is received by the load. When the load is actually connected, some combination of current and voltage will result and power will be delivered. To figure out how much power, we have to consider the $I-V$ characteristic curve of the module as well as the $I-V$ characteristic curve of the load.

Figure 8.34 shows a generic $I-V$ curve for a PV module, identifying several key parameters including the open-circuit voltage V_{OC} and the short-circuit current I_{SC}. Also shown is the product of voltage and current, that is, power delivered by the module. At the two ends of the $I-V$ curve, the output power is zero since either current or voltage is zero at those points. The maximum power point (MPP) is that spot near the knee of the $I-V$ curve at which the product of current and voltage reaches its maximum. The voltage and current at the MPP are sometimes designated as V_m and I_m for the general case and designated V_R and I_R (for *rated voltage* and *rated current*) under the special circumstances that correspond to idealized test conditions.

Another way to visualize the location of the maximum power point is by imagining trying to find the biggest possible rectangle that will fit beneath the $I-V$ curve. As shown in Fig. 8.35, the sides of the rectangle correspond to current and voltage, so its area is power. Another quantity that is often used to characterize module performance is the *fill factor* (FF). The fill factor is the ratio of the power at the maximum power point to the product of V_{OC} and I_{SC}, so FF can be visualized as the ratio of two rectangular areas, as is suggested in

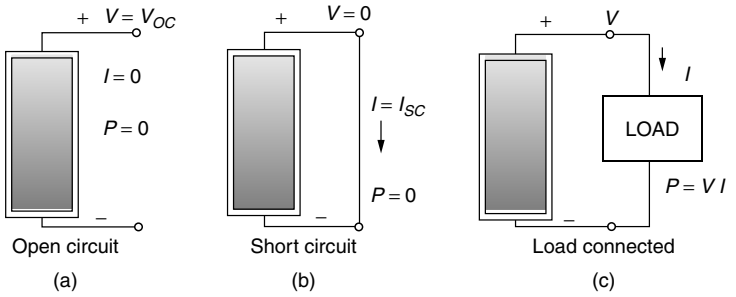


Figure 8.33 No power is delivered when the circuit is open (a) or shorted (b). When the load is connected (c), the same current flows through the load and module and the same voltage appears across them.

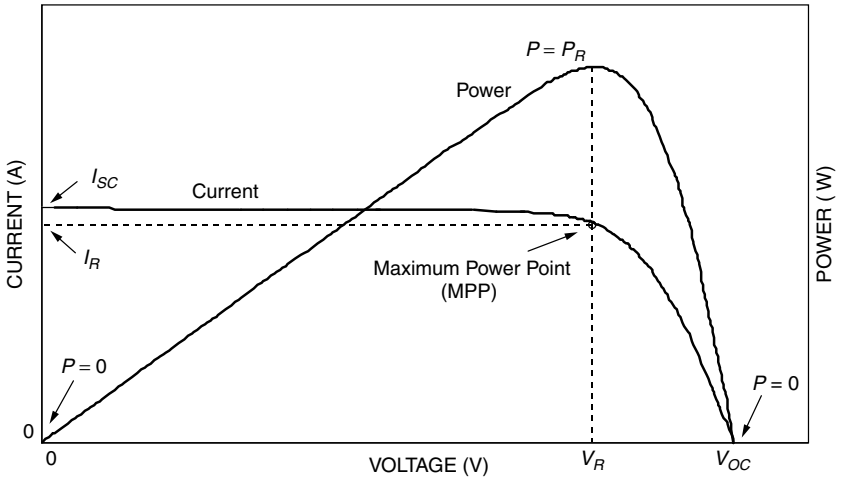


Figure 8.34 The $I-V$ curve and power output for a PV module. At the maximum power point (MPP) the module delivers the most power that it can under the conditions of sunlight and temperature for which the $I-V$ curve has been drawn.

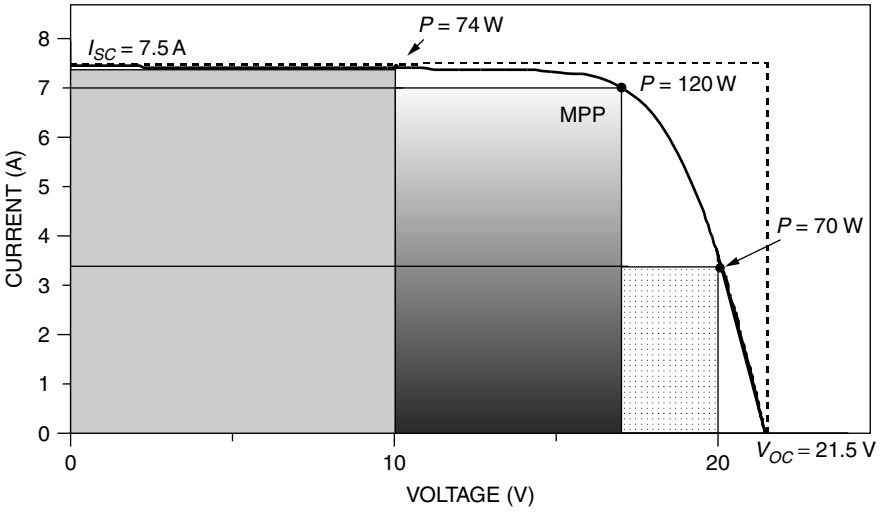


Figure 8.35 The maximum power point (MPP) corresponds to the biggest rectangle that can fit beneath the $I-V$ curve. The fill factor (FF) is the ratio of the area (power) at MPP to the area formed by a rectangle with sides V_{OC} and I_{SC}.

Fig. 8.35. Fill factors around 70–75% for crystalline silicon solar modules are typical, while for multijunction amorphous-Si modules, it is closer to 50–60%.

$$\text{Fill factor (FF)} = \frac{\text{Power at the maximum power point}}{V_{OC}\ I_{SC}} = \frac{V_R I_R}{V_{OC} I_{SC}} \qquad (8.23)$$

TABLE 8.3 Examples of PV Module Performance Data Under Standard Test Conditions (1 kW/m^2, AM 1.5, 25°C Cell Temperature)

Manufacturer	Kyocera	Sharp	BP	Uni-Solar	Shell
Model	KC-120-1	NE-Q5E2U	2150S	US-64	ST40
Material	Multicrystal	Polycrystal	Monocrystal	Triple junction a-Si	CIS-thin film
Number of cells n	36	72	72		42
Rated Power $P_{DC,STC}$ (W)	120	165	150	64	40
Voltage at max power (V)	16.9	34.6	34	16.5	16.6
Current at rated power (A)	7.1	4.77	4.45	3.88	2.41
Open-circuit voltage V_{OC} (V)	21.5	43.1	42.8	23.8	23.3
Short-circuit current I_{SC} (A)	7.45	5.46	4.75	4.80	2.68
Length (mm/in.)	1425/56.1	1575/62.05	1587/62.5	1366/53.78	1293/50.9
Width (mm/in.)	652/25.7	826/32.44	790/31.1	741/29.18	329/12.9
Depth (mm/in.)	52/2.0	46/1.81	50/1.97	31.8/1.25	54/2.1
Weight (kg/lb)	11.9/26.3	17/37.5	15.4/34	9.2/20.2	14.8/32.6
Module efficiency	12.9%	12.7%	12.0%	6.3%	9.4%

Since PV $I-V$ curves shift all around as the amount of insolation changes and as the temperature of the cells varies, standard test conditions (STC) have been established to enable fair comparisons of one module to another. Those test conditions include a solar irradiance of 1 kW/m^2 (1 sun) with spectral distribution shown in Fig. 8.10, corresponding to an air mass ratio of 1.5 (AM 1.5). The standard cell temperature for testing purposes is 25°C (it is important to note that 25° is cell temperature, not ambient temperature). Manufacturers always provide performance data under these operating conditions, some examples of which are shown in Table 8.3. The key parameter for a module is its rated power; to help us remember that it is dc power measured under standard test conditions, it has been identified in Table 8.3 as $P_{DC,STC}$. Later we'll learn how to adjust rated power to account for temperature effects as well as see how to adjust it to give us an estimate of the actual ac power that the module and inverter combination will deliver.

8.6 IMPACTS OF TEMPERATURE AND INSOLATION ON *I–V* CURVES

Manufacturers will often provide $I-V$ curves that show how the curves shift as insolation and cell temperature changes. Figure 8.36 shows examples for the Kyocera 120-W multicrystal-silicon module described in Table 8.3. Notice as insolation drops, short-circuit current drops in direct proportion. Cutting insolation in half, for example, drops I_{SC} by half. Decreasing insolation also reduces

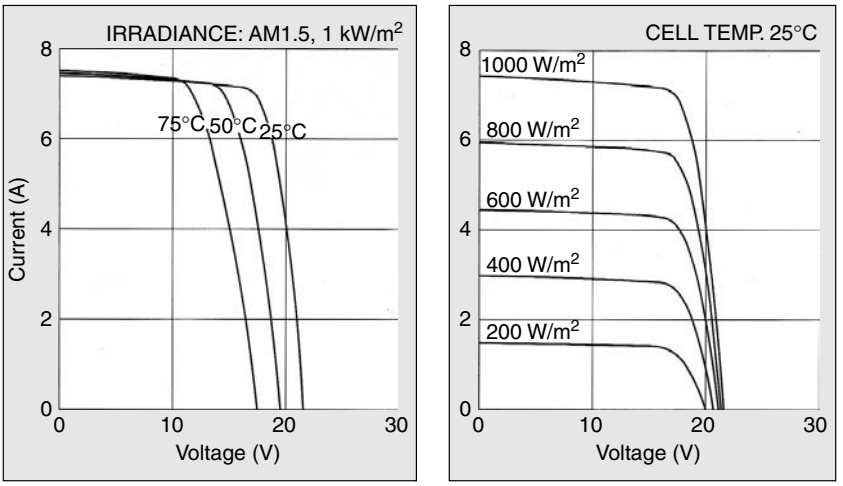


Figure 8.36 Current-voltage characteristic curves under various cell temperatures and irradiance levels for the Kyocera KC120-1 PV module.

V_{OC}, but it does so following a logarithmic relationship that results in relatively modest changes in V_{OC}.

As can be seen in Fig. 8.36, as cell temperature increases, the open-circuit voltage decreases substantially while the short-circuit current increases only slightly. Photovoltaics, perhaps surprisingly, therefore perform better on cold, clear days than hot ones. For crystalline silicon cells, V_{OC} drops by about 0.37% for each degree Celsius increase in temperature and I_{SC} increases by approximately 0.05%. The net result when cells heat up is the MPP slides slightly upward and toward the left with a decrease in maximum power available of about 0.5%/°C. Given this significant shift in performance as cell temperature changes, it should be quite apparent that temperature needs to be included in any estimate of module performance.

Cells vary in temperature not only because ambient temperatures change, but also because insolation on the cells changes. Since only a small fraction of the insolation hitting a module is converted to electricity and carried away, most of that incident energy is absorbed and converted to heat. To help system designers account for changes in cell performance with temperature, manufacturers often provide an indicator called the NOCT, which stands for nominal operating cell temperature. The NOCT is cell temperature in a module when ambient is 20°C, solar irradiation is 0.8 kW/m^2, and windspeed is 1 m/s. To account for other ambient conditions, the following expression may be used:

$$T_{\text{cell}} = T_{\text{amb}} + \left(\frac{\text{NOCT} - 20^{\circ}}{0.8}\right) \cdot S \tag{8.24}$$

where T_{cell} is cell temperature (°C), T_{amb} is ambient temperature, and S is solar insolation (kW/m^2).

Example 8.5 Impact of Cell Temperature on Power for a PV Module. Estimate cell temperature, open-circuit voltage, and maximum power output for the 150-W BP2150S module under conditions of 1-sun insolation and ambient temperature 30°C. The module has a NOCT of 47°C.

Solution. Using (8.24) with $S = 1$ kW/m^2, cell temperature is estimated to be

$$T_{\text{cell}} = T_{\text{amb}} + \left(\frac{\text{NOCT} - 20^\circ}{0.8}\right) \cdot S = 30 + \left(\frac{47 - 20}{0.8}\right) \cdot 1 = 64^\circ\text{C}$$

From Table 8.3, for this module at the standard temperature of 25°C, $V_{OC} =$ 42.8 V. Since V_{OC} drops by 0.37%/°C, the new V_{OC} will be about

$$V_{OC} = 42.8[1 - 0.0037(64 - 25)] = 36.7 \text{ V}$$

With maximum power expected to drop about 0.5%/°C, this 150-W module at its maximum power point will deliver

$$P_{\text{max}} = 150 \text{ W} \cdot [1 - 0.005(64 - 25)] = 121 \text{ W}$$

which is a rather significant drop of 19% from its rated power.

When the NOCT is not given, another approach to estimating cell temperature is based on the following:

$$T_{\text{cell}} = T_{\text{amb}} + \gamma \left(\frac{\text{Insolation}}{1 \text{ kW/m}^2}\right) \tag{8.25}$$

where γ is a proportionality factor that depends somewhat on windspeed and how well ventilated the modules are when installed. Typical values of γ range between 25°C and 35°C; that is, in 1 sun of insolation, cells tend to be 25–35°C hotter than their environment.

8.7 SHADING IMPACTS ON I–V CURVES

The output of a PV module can be reduced dramatically when even a small portion of it is shaded. Unless special efforts are made to compensate for shade problems, even a single shaded cell in a long string of cells can easily cut

output power by more than half. External diodes, purposely added by the PV manufacturer or by the system designer, can help preserve the performance of PV modules. The main purpose for such diodes is to mitigate the impacts of shading on PV $I-V$ curves. Such diodes are usually added in parallel with modules or blocks of cells within a module.

8.7.1 Physics of Shading

To help understand this important shading phenomenon, consider Fig. 8.37 in which an n-cell module with current I and output voltage V shows one cell separated from the others (shown as the top cell, though it can be any cell in the string). The equivalent circuit of the top cell has been drawn using Fig. 8.26, while the other $(n-1)$ cells in the string are shown as just a module with current I and output voltage V_{n-1}.

In Fig. 8.37a, all of the cells are in the sun and since they are in series, the same current I flows through each of them. In Fig. 8.37b, however, the top cell is shaded and its current source I_{SC} has been reduced to zero. The voltage drop across R_P as current flows through it causes the diode to be reverse biased, so the diode current is also (essentially) zero. That means the entire current flowing through the module must travel through both R_P and R_S in the shaded cell on its way to the load. That means the top cell, instead of adding to the output voltage, actually reduces it.

Consider the case when the bottom $n-1$ cells still have full sun and still some how carry their original current I so they will still produce their original voltage V_{n-1}. This means that the output voltage of the entire module V_{SH} with

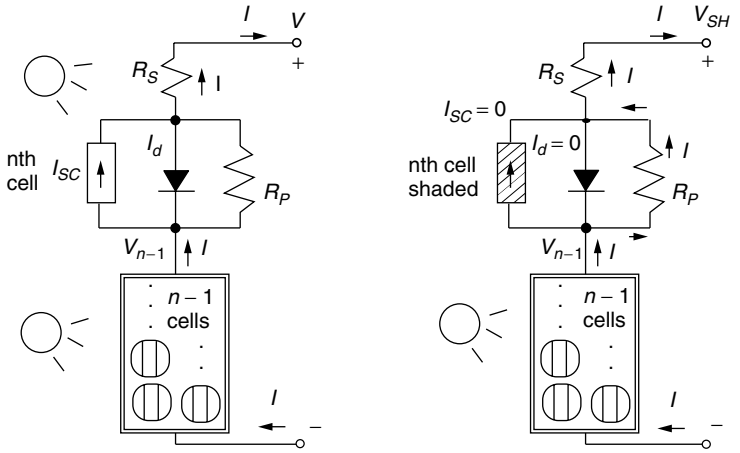


Figure 8.37 A module with n cells in which the top cell is in the sun (a) or in the shade (b).

one cell shaded will drop to

$$V_{SH} = V_{n-1} - I(R_P + R_S) \tag{8.26}$$

With all n cells in the sun and carrying I, the output voltage was V so the voltage of the bottom $n - 1$ cells will be

$$V_{n-1} = \left(\frac{n-1}{n}\right) V \tag{8.27}$$

Combining (8.26) and (8.27) gives

$$V_{SH} = \left(\frac{n-1}{n}\right) V - I(R_P + R_S) \tag{8.28}$$

The drop in voltage Δ V at any given current I, caused by the shaded cell, is given by

$$\Delta V = V - V_{SH} = V - \left(1 - \frac{1}{n}\right) V + I(R_P + R_S) \tag{8.29}$$

$$\Delta V = \frac{V}{n} + I(R_P + R_S) \tag{8.30}$$

Since the parallel resistance R_P is so much greater than the series resistance R_S, (8.30) simplifies to

$$\Delta V \cong \frac{V}{n} + IR_P \tag{8.31}$$

At any given current, the $I-V$ curve for the module with one shaded cell drops by ΔV. The huge impact this can have is illustrated in Fig. 8.38.

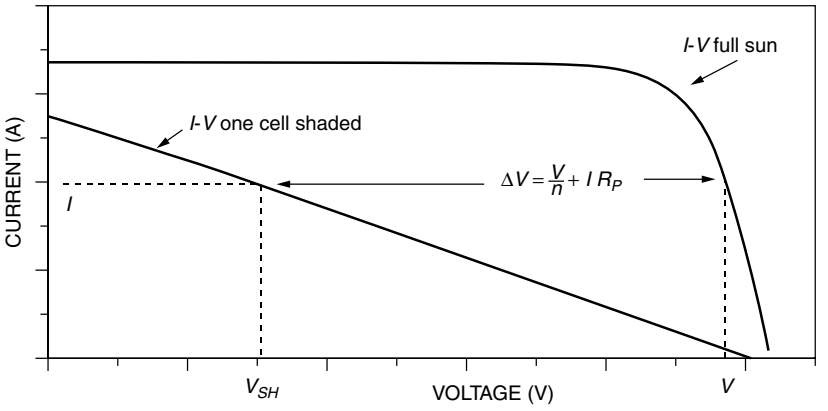


Figure 8.38 Effect of shading one cell in an n-cell module. At any given current, module voltage drops from V to $V - \Delta V$.

Example 8.6 Impacts of Shading on a PV Module. The 36-cell PV module described in Example 8.4 had a parallel resistance per cell of $R_P = 6.6\ \Omega$. In full sun and at current $I = 2.14$ A the output voltage was found there to be $V = 19.41$ V. If one cell is shaded and this current somehow stays the same, then:

a. What would be the new module output voltage and power?
b. What would be the voltage drop across the shaded cell?
c. How much power would be dissipated in the shaded cell?

Solution

a. From (8.31) the drop in module voltage will be

$$\Delta V = \frac{V}{n} + I R_P = \frac{19.41}{36} + 2.14 \times 6.6 = 14.66 \text{ V}$$

The new output voltage will be $19.41 - 14.66 = 4.75$ V.
Power delivered by the module with one cell shaded would be

$$P_{\text{module}} = V I = 4.75 \text{ V} \times 2.14 \text{ A} = 10.1 \text{ W}$$

For comparison, in full sun the module was producing 41.5 W.

b. All of that 2.14 A of current goes through the parallel plus series resistance (0.005 Ω) of the shaded cell, so the drop across the shaded cell will be

$$V_c = I(R_P + R_S) = 2.14(6.6 + 0.005) = 14.14\ V$$

(normally a cell in the sun will add about 0.5 V to the module; this shaded cell subtracts over 14 V from the module).

c. The power dissipated in the shaded cell is voltage drop times current, which is

$$P = V_c I = 14.14 \text{ V} \times 2.14 \text{ A} = 30.2 \text{ W}$$

All of that power dissipated in the shaded cell is converted to heat, which can cause a local hot spot that may permanently damage the plastic laminates enclosing the cell.

The procedures demonstrated in Examples 8.4 and 8.6 can be extended to develop I–V curves under various conditions of shading. Figure 8.39 shows such curves for the example module under full-sun conditions and with one cell 50% shaded, one cell completely shaded, and two cells completely shaded. Also shown on the graph is a dashed vertical line at 13 V, which is a typical operating

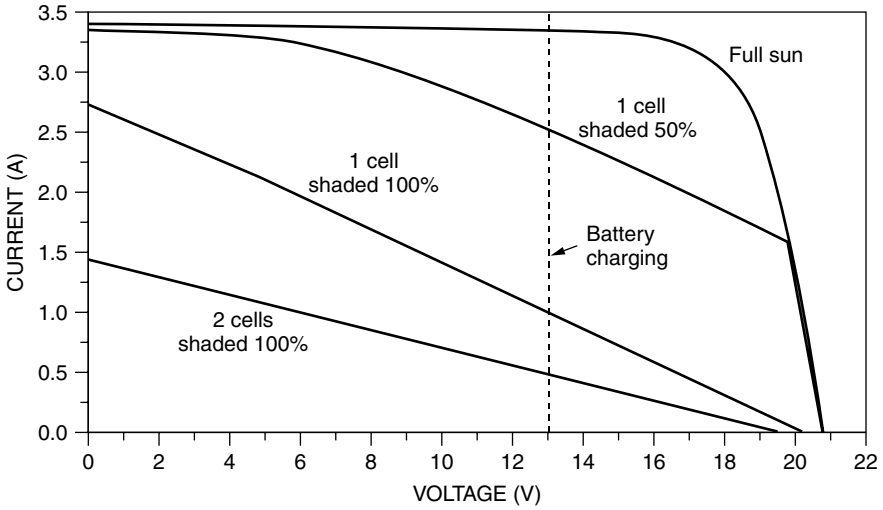


Figure 8.39 Effects of shading on the $I-V$ curves for a PV module. The dashed line shows a typical voltage that the module would operate at when charging a 12-V battery; the impact on charging current is obviously severe.

voltage for a module charging a 12-V battery. The reduction in charging current for even modest amounts of shading is severe. With just one cell shaded out of 36 in the module, the power delivered to the battery is decreased by about two-thirds!

8.7.2 Bypass Diodes for Shade Mitigation

Example 8.6 shows not only how drastically shading can shift the $I-V$ curve, but also how local, potentially damaging hot spots can be created in shaded cells. Figure 8.40 shows a typical situation. In Fig. 8.40a a solar cell in full sun operating in its normal range *contributes* about 0.5 V to the voltage output of the module, but in the equivalent circuit shown in 8.40b a shaded cell experiences a *drop* as current is diverted through the parallel and series resistances. This drop can be considerable (in Example 8.6 it was over 14 V).

The voltage drop problem in shaded cells could be to corrected by adding a *bypass diode* across each cell, as shown in Fig. 8.41. When a solar cell is in the sun, there is a voltage rise across the cell so the bypass diode is cut off and no current flows through it—it is as if the diode is not even there. When the solar cell is shaded, however, the drop that would occur if the cell conducted any current would turn on the bypass diode, diverting the current flow through that diode. The bypass diode, when it conducts, drops about 0.6 V. So, the bypass diode controls the voltage drop across the shaded cell, limiting it to a relatively modest 0.6 V instead of the rather large drop that may occur without it.

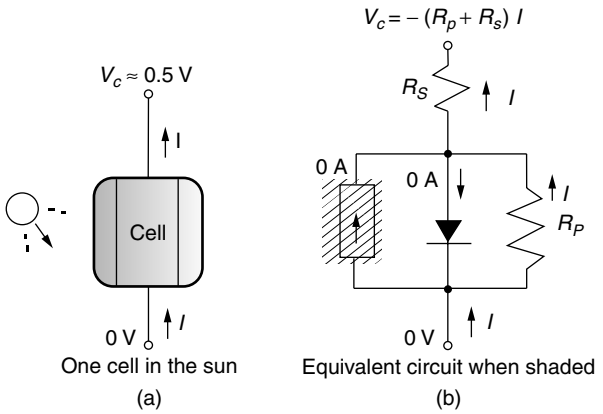


Figure 8.40 In full sun a cell may contribute around 0.5 V to the module output; but when a cell is shaded, it can have a large voltage drop across it.

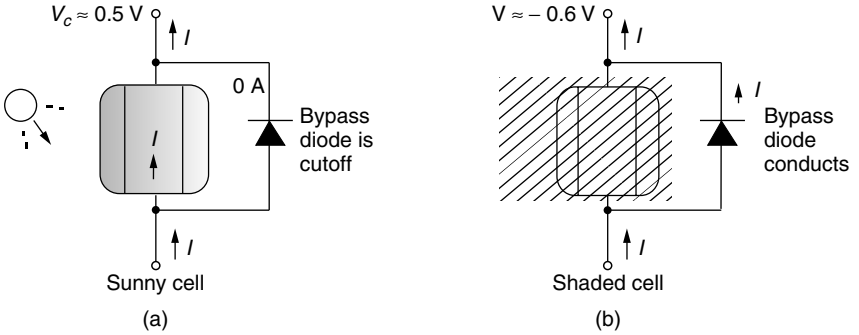


Figure 8.41 Mitigating the shade problem with a bypass diode. In the sun (a), the bypass diode is cut off and all the normal current goes through the solar cell. In shade (b), the bypass diode conducts current around the shaded cell, allowing just the diode drop of about 0.6 V to occur.

In real modules, it would be impractical to add bypass diodes across every solar cell, but manufacturers often do provide at least one bypass diode around a module to help protect arrays, and sometimes several such diodes around groups of cells within a module. These diodes don't have much impact on shading problems of a single module, but they can be very important when a number of modules are connected in series. Just as cells are wired in series to increase module voltage, modules can be wired in series to increase array voltage. Also, just as a single cell can drag down the current within a module, a few shaded cells in a single module can drag down the current delivered by the entire string in an array. The benefit already demonstrated for a bypass diode on a single cell also applies to a diode applied across a complete module.

To see how bypass diodes wired in parallel with modules can help mitigate shading problems, consider Fig. 8.42, which shows $I-V$ curves for a string of five modules (the same modules that were used to derive Fig. 8.39). The graph shows the modules in full sun as well as the $I-V$ curve that results when one module has two cells completely shaded. Imagine the PVs delivering charging current at about 65 V to a 60-V battery bank. As can be seen, in full sun about 3.3 A are delivered to the batteries. However, when just two cells in one module are shaded, the current drops by one-third to about 2.2 A. With a bypass diode across the shaded module, however, the $I-V$ curve is improved considerably as shown in the figure.

Figure 8.43 helps explain how the bypass diodes do their job. Imagine five modules, wired in series, connected to a battery that forces the modules to operate at 65 V. In full sun the modules deliver 3.3 A at 65 V. When any of the cells are shaded, they cease to produce voltage and instead begin to act like resistors (6.6 Ω per cell in this example) that cause voltage to drop as the other modules continue to try to push current through the string. Without a bypass diode to divert the current, the shaded module loses voltage and the other modules try to compensate by increasing voltage, but the net effect is that current in the whole string drops. If, however, bypass diodes are provided, as shown in Fig. 8.43c, then current will go around the shaded module and the charging current bounces back to nearly the same level that it was before shading occurred.

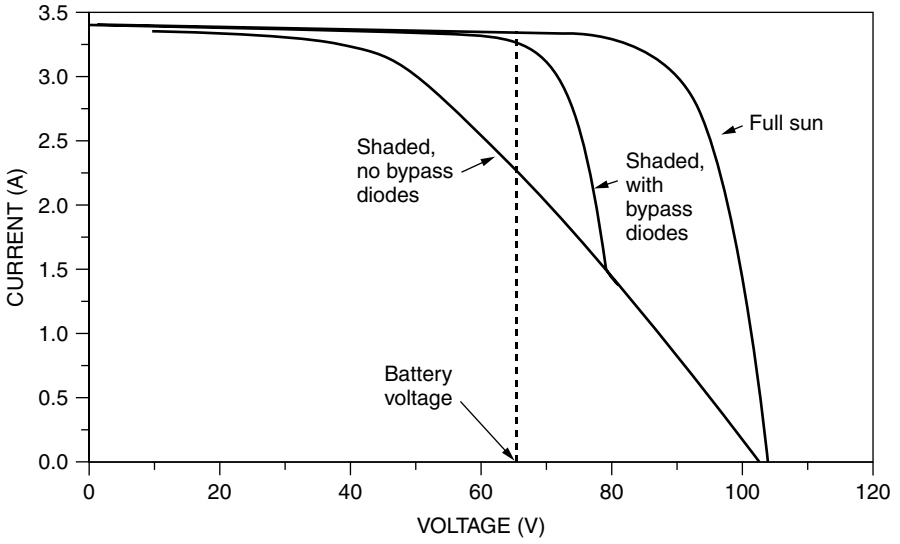


Figure 8.42 Impact of bypass diodes. Drawn for five modules in series delivering 65 V to a battery bank. With one module having two shaded cells, charging current drops by almost one-third when there are no bypass diodes. With the module bypass diodes there is very little drop.

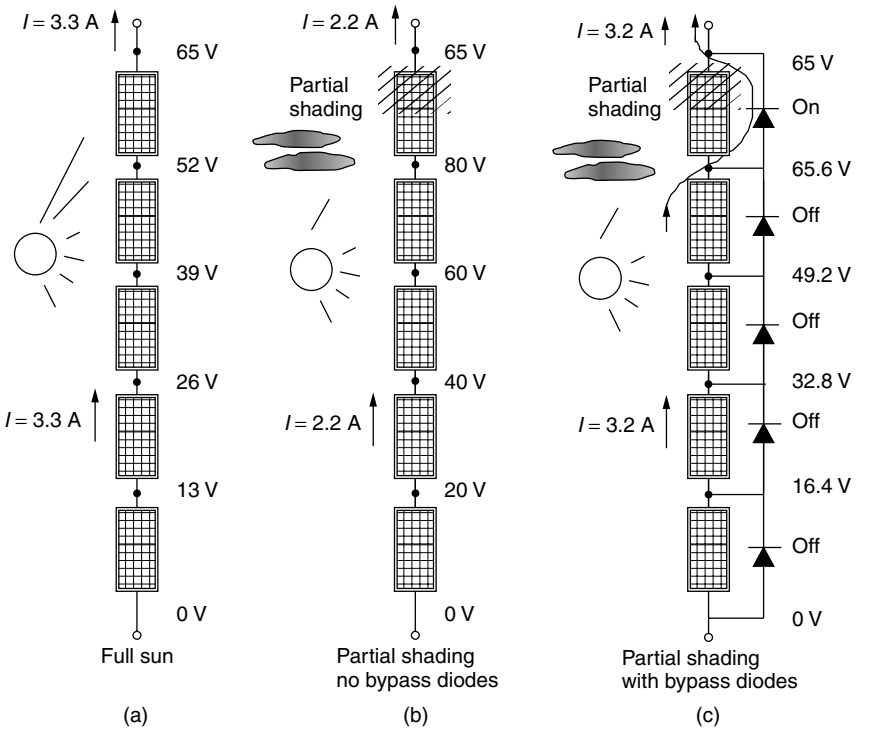


Figure 8.43 Showing the ability of bypass diodes to mitigate shading when modules are charging a 65 V battery. Without bypass diodes, a partially shaded module constricts the current delivered to the load (b). With bypass diodes, current is diverted around the shaded module.

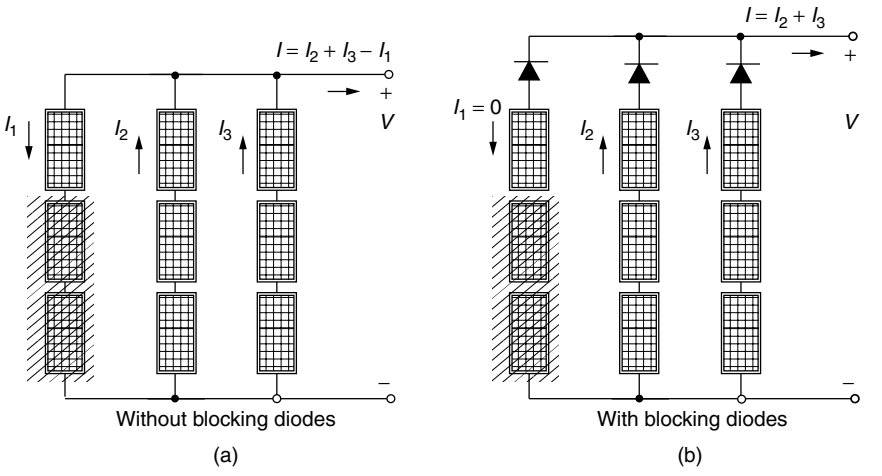


Figure 8.44 Blocking diodes prevent reverse current from flowing down malfunctioning or shaded strings.

8.7.3 Blocking Diodes

Bypass diodes help current go around a shaded or malfunctioning module within a string. This not only improves the string performance, but also prevents hot spots from developing in individual shaded cells. When strings of modules are wired in parallel, a similar problem may arise when one of the strings is not performing well. Instead of supplying current to the array, a malfunctioning or shaded string can withdraw current from the rest of the array. By placing *blocking diodes* (also called *isolation diodes*) at the top of each string as shown in Fig. 8.44, the reverse current drawn by a shaded string can be prevented.

8.8 CRYSTALLINE SILICON TECHNOLOGIES

Thus far, the discussion of photovoltaics has been quite generic; that is, it hasn't particularly depended upon what technology was used to fabricate the cells. The circuit concepts just developed will be used in the next chapter when we explore PV system design, but before we get there it will be helpful to explore the different types of technologies currently used to manufacture photovoltaics.

There are a number of ways to categorize photovoltaics. One dichotomy is based on the thickness of the semiconductor. Conventional crystalline silicon solar cells are, relatively speaking, very thick—on the order of 200–500 μm (0.008–0.020 in.). An alternative approach to PV fabrication is based on thin films of semiconductor, where "thin" means something like 1–10 μm. Thin-film cells require much less semiconductor material and are easier to manufacture, so they have the potential to be cheaper than thick cells. The first generation of thin-film PVs were only about half as efficient as conventional thick silicon cells; they were less reliable over time, yet they were no cheaper per watt, so they really weren't competitive. All three of these negative attributes have been addressed, more or less successfully, and thin-film PVs are beginning to become more competitive. In the near future they may even dominate PV sales. Currently, however, about 80% of all photovoltaics are thick cells and the remaining 20% are thin-film cells used mostly in calculators, watches, and other consumer electronics.

Photovoltaic technologies can also be categorized by the extent to which atoms bond with each other in individual crystals. As described by Bube (1998), there is a "family tree" of PVs based on the size of these crystals. The historically generic name "polycrystalline" can be broken down into the following more specific terms: (1) *single crystal*, the dominant silicon technology; (2) *multicrystalline*, in which the cell is made up of a number of relatively large areas of single crystal grains, each on the order of 1 mm to 10 cm in size, including multicrystalline silicon (mc-Si); (3) *polycrystalline*, with many grains having dimensions on the order of 1 μm to 1 mm, as is the case for cadmium telluride (CdTe) cells, copper indium diselenide ($CuInSe_2$,) and polycrystalline, thin-film silicon; (4) *microcrystalline* cells with grain sizes less than 1 μm; and (5) *amorphous*, in which there are no single-crystal regions, as in amorphous silicon (a-Si).

Another way to categorize photovoltaic materials is based on whether the p and n regions of the semiconductor are made of the same material (with different dopings, of course)—for example, silicon. These are called *homojunction* photovoltaics. When the p–n junction is formed between two different semiconductors, they are called *heterojunction* PVs. For example, one of the most promising heterojunction combinations uses cadmium sulfide (CdS) for the n-type layer and copper indium diselenide ($CuInSe_2$, also known as "CIS") for the p-type layer.

Other distinctions include multiple junction solar cells (also known as cascade or tandem cells) made up of a stack of p–n junctions with each junction designed to capture a different portion of the solar spectrum. The shortest-wavelength, highest-energy photons are captured in the top layer while most of the rest pass through to the next layer. Subsequent layers have lower and lower band gaps, so they each pick off the most energetic photons that they see, while passing the rest down to the next layer. Very high efficiencies are possible using this approach.

Finally, some cells are specifically designed to work best with concentrated sunlight while others are used in nonconcentrating flat-plate systems. Figure 8.45 provides a rough road map for our subsequent discussion of each technology.

8.8.1 Single-Crystal Czochralski (CZ) Silicon

Silicon is the second most abundant element on earth, comprising approximately 20% of the earth's crust. Pure silicon almost immediately forms a layer of SiO_2 on its surface when exposed to air, so it exists in nature mostly in SiO_2-based minerals such as quartzite or in silicates such as mica, feldspars, and zeolites.

The raw material for photovoltaics and other semiconductors could be common sand, but it is usually naturally purified, high-quality silica or quartz (SiO_2) from mines. The first processing step to convert silica into crystalline silicon is an energy-intensive process in which SiO_2 is reduced to an impure Si using carbon in an arc furnace. It is then transformed to a liquid trichlorosilane ($SiHCl_3$),

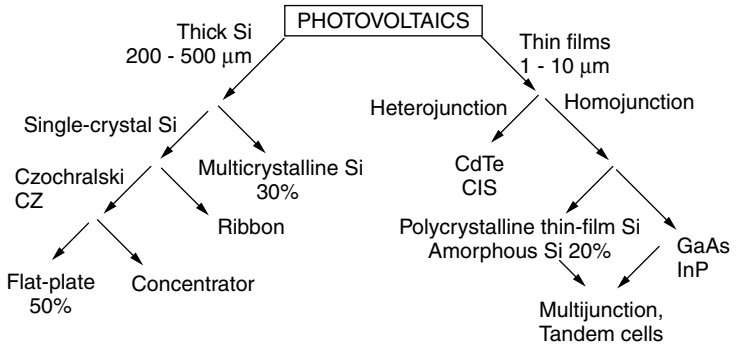


Figure 8.45 One way to organize the discussion of photovoltaic technologies. Percentages represent fraction of PV sales in the late 1990s.

which can be purified by fractional distillation. The purified trichlorosilane can then be reduced with hydrogen to form extremely pure silicon:

$$SiHCl_3 + H_2 + \text{heat} \rightarrow Si + 3HCl \tag{8.32}$$

The result is rock-like chunks of 99.9999% pure silicon. When heated to over 1400°C, the rocks can be melted in a quartz crucible to form a molten vat of silicon.

The most commonly used technique for forming single-crystal silicon from the crucible of molten silicon is the *Czochralski*, or CZ, method (Fig. 8.46a), in which a small seed of solid, crystalline silicon about the size of a pencil is dipped into the vat and then slowly withdrawn using a combination of pulling (10^{-4}–10^{-2} cm/s) and rotating (10–40 rpm). As it is withdrawn, the molten silicon atoms bond with atoms in the crystal and then solidify (freeze) in place. The result is a large cylindrical ingot or "boule" of single-crystal silicon perhaps a meter long and as large as about 20 cm in diameter. By adding proper amounts of a dopant to the melt, the resulting ingot can be fabricated as an *n*- or *p*-type material. Usually the dopant is boron and the ingot is therefore a *p*-type semiconductor. An alternative to the Czochralski method is called the *float-zone* (FZ) process, in which a solid ingot of silicon is locally melted and then solidified by an RF field that passes slowly along the ingot.

After the cylindrical ingot is formed, four sides may be sliced off, resulting in a block of silicon that has an approximately square cross section. This "square-ness" allows greater packing density when the cells are assembled into a photovoltaic module. The silicon block must then be sawed into thin slices called *wafers*. This can be done with a saw blade (Fig. 8.46b) or with a diamond impregnated wire that cuts through the wafer. In either case, a large fraction of the ingot is lost as kerf (like sawdust)—as much as 50% when a saw blade is used, less if the wire saw is used. The wafers are then etched to remove some of the surface damage and to expose the microscopic crystalline structure at the top of the cell. The surface is made up of a jumble of four-sided pyramids, which

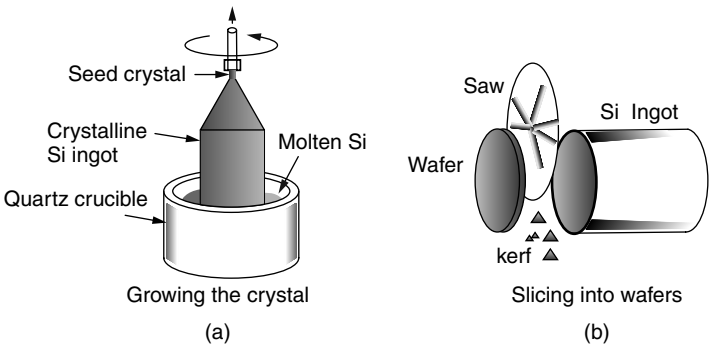


Figure 8.46 The Czochralski method for growing single-crystal silicon.

helps reflect light down into the crystal. After polishing, the wafers are ready to be doped to make the $p-n$ junction.

During the above wafer fabrication, the crystalline silicon is usually doped with acceptor atoms, making it p-type throughout its 200- to- 500-μm thickness. To form the junction, a thin 0.1- to 0.5-μm n-type layer is created by diffusing enough donor atoms into the top of the cell to overwhelm the already existing acceptors. The wafers are placed in long tubes of silica glass for the diffusion process. The impurities, in gaseous form, flow through the tubes, thereby exposing the wafers under carefully controlled exposure time and temperature conditions. For most crystalline silicon, the donor atoms are phosphorus from phosphine gas (PH_3) and the acceptors are boron (from diborane, B_2H_6).

Since silicon is naturally quite reflective to solar wavelengths, some sort of surface treatment is required to reduce those losses. An antireflection (AR) coating of some transparent material such as tin oxide (SnO_2) is applied. These coatings tend to readily transmit the green, yellow, and red light into the cell, but some of the shorter-wavelength blue light is reflected, which gives the cells their characteristic dark blue color.

The next step is the attachment of electrical contacts to the cell. For many years, the bottom contacts were formed by vacuum deposition of a layer of aluminum that covered the back side of the cell. Aluminum is a Group III element, so it not only serves as a conductor but also can contribute to the concentration of holes in the bottom of the p-layer, forming what is called a p^+ layer. Those extra holes help reduce the contact resistance between the silicon and aluminum, and the gradient of holes that they create helps reduce recombination at the contact by driving holes away from the back surface. A cross section of a typical 1970s vintage, single-crystal silicon cell is shown in Fig. 8.47a.

The front-surface contacts in most cells have been formed by depositing a grid of metal conductors that covers on the order of 5–10% of the total area. That coverage, of course, reduces the amount of sunlight reaching the junction and hence reduces the overall cell efficiency. Some newer cells, called *back-point contact cells*, put both contacts on the bottom to avoid that shading effect. Another approach involves use of lasers to dig deep, narrow grooves into the cell. The deep grooves in these *laser-grooved, buried-contact cells* are filled with metal, forming a large contact area while minimizing the top-surface area shaded by the contact. The bottom contacts can also be formed using this laser technique, resulting in what is called a "double-sided" laser-grooved photovoltaic cell (Fig. 8.47b).

In newer cells, a number of other techniques may be incorporated into the top surface to improve performance. One of the most advanced crystalline silicon photovoltaics is called *the passive-emitter, rear locally diffused* (PERL) cell (Fig. 8.47c). In a PERL cell, inverted pyramids on the front surface, covered with an antireflection coating, help capture and bounce light into the cell. These cells not only direct more sunlight into the cell, they also reflect back into the cell photons that were reflected off of the bottom oxide layer covered by aluminum. Efficiencies approaching 25% have been achieved.

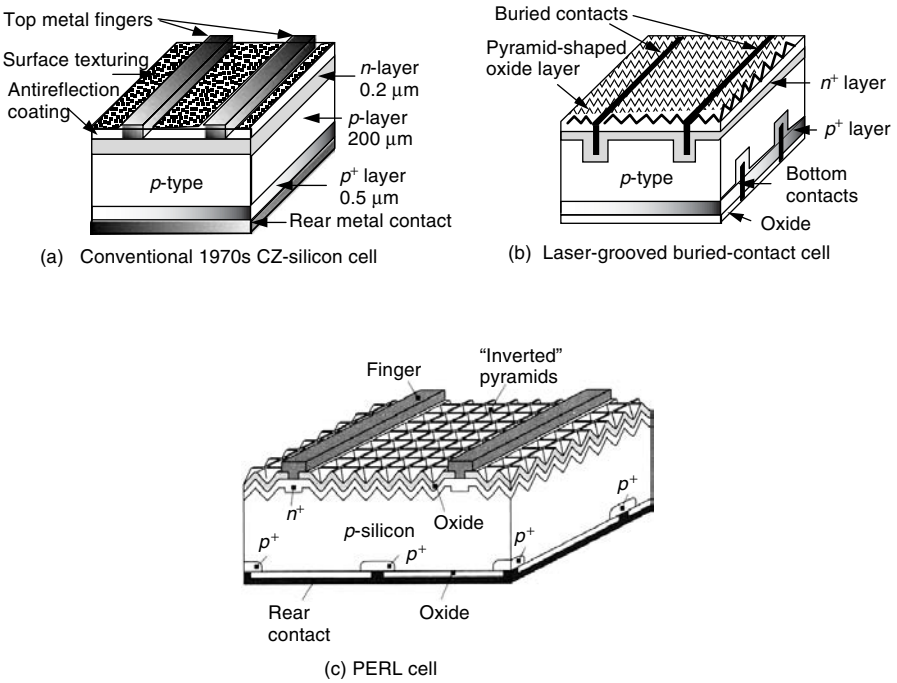


Figure 8.47 Evolution of CZ-silicon solar cells. (a) A conventional 1970s cell, showing typical thicknesses of semiconductor. (b) Double-sided, laser-grooved, buried-contact cell. (c) Passive-emitter, rear locally diffused (PERL) cell. Based on Green (1993).

Great progress has been made in improving the maximum efficiency achieved in laboratory cells. Figure 8.48 illustrates the rise from less than 1% in the 1940s, to just over 15% in the early 1970s, to over 25% by 2000. Actual production module efficiencies in 2003 were in the range of 14 to 17%.

8.8.2 Ribbon Silicon Technologies

The fact that the CZ and FZ processes for wafer production were developed for the semiconductor industry has been both a blessing and a curse. Without question, the photovoltaics industry has benefited greatly by being able to piggyback onto technology developed for the immense microelectronics industry. Also, since PVs don't need the very highest quality wafers, they can use some of the rejected crystalline silicon originally meant for the semiconductor industry. On the other hand, a wafer that is fabricated into thousands of high-priced integrated circuits can generate a lot more revenue than one that ends up as a photovoltaic cell, which means an electronics company can pay a lot more for wafers, keeping their price high.

With the cost of sliced-and-polished wafers being a significant fraction of the cost of PVs, attempts have been made to find other ways to fabricate crystalline

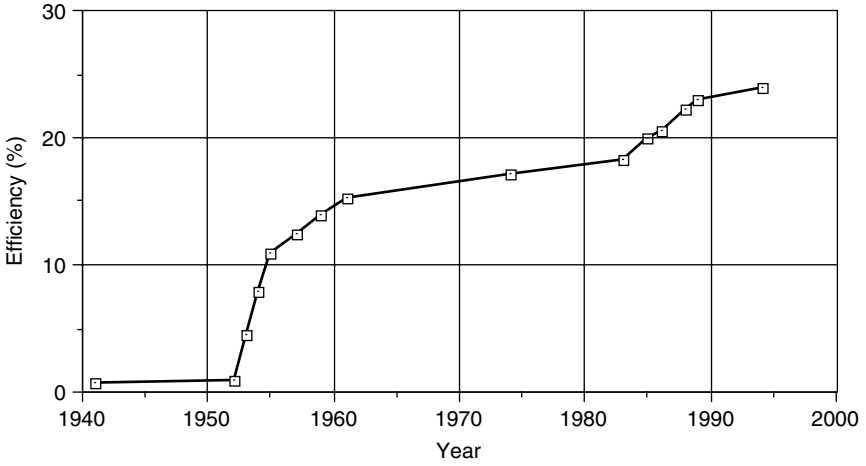


Figure 8.48 Increasing efficiency of single-crystal silicon, laboratory-scale cells. From Bube (1998).

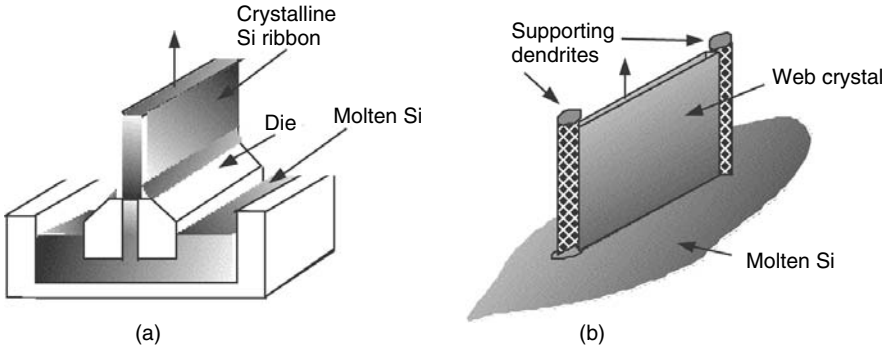


Figure 8.49 Ribbons of crystalline silicon can be grown using the edge-defined film-fed growth process (EFG) in (a) or using the dendritic web process (b).

silicon. Several such technologies are based on growing crystalline silicon that emerges as a long, thin, continuous ribbon from the silicon melt. The ribbons can then be scribed and broken into rectangular cells without the wastefulness of sawing an ingot and without the need for separate polishing steps.

One such approach, called "edge-defined, film-fed growth (EFG)," is illustrated in Fig. 8.49a. It is based on a carbon die that is partially immersed in molten silicon. Within the slot-shaped aperture in the die, molten silicon solidifies and emerges as a frozen ribbon of crystalline silicon. An earlier ribbon process was based on growth of a thin sheet of crystal between two parallel dendrite supports (Fig. 8.49b). Very precise temperature control is required for the dendrites, and the solidifying silicon between them, to form good crystalline material as

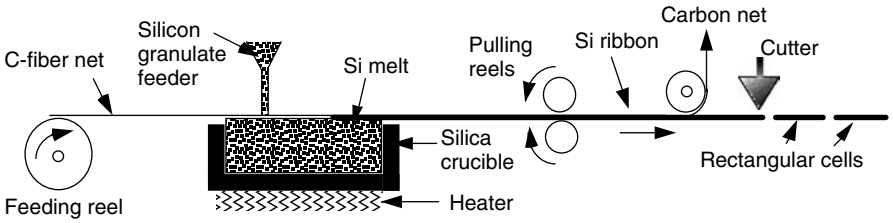


Figure 8.50 The S-Web process produces a continuous ribbon of silicon, which can be doped and cut into individual rectangular cells. Based on Schmela (2000).

the ribbon is pulled from the melt. A similar process is called the String Ribbon technique, in which two high-temperature vertical strings are pulled vertically through a shallow silicon melt. The molten silicon spans and freezes between the strings, forming a long, thin single-crystal ribbon. The silicon ribbon is then cut to length without interrupting the continuous growth of new ribbon.

A third ribbon technology was introduced by Siemens Solar in 1990, but was never fully developed until the German company, Brandl AG, picked it up in 2000. The S-Web process utilizes a carbon-fiber net that is pulled horizontally along the surface of a silicon melt. The silicon solidifies along the lower side of the net, forming a crystal that can be drawn out into a continuous ribbon, which can then be doped and cut into rectangular, cell-sized pieces. A sketch of the process is shown in Fig. 8.50.

8.8.3 Cast Multicrystalline Silicon

Another way to avoid the costly Czochralski and float-zone processes is based on carefully cooling and solidifying a crucible of molten silicon, yielding a large, solid rectangular ingot. Since these ingots may be quite large, on the order of $40 \times 40 \times 40$ cm and weighing over 100 kg, the ingot may need to be cut into smaller, more manageable blocks, which are then sliced into silicon wafers using either the saw or wire-cutting techniques. Sawing can waste a significant fraction of the ingot, but since this casting method is itself cheaper and it utilizes less expensive, less pure silicon than the CZ process, the waste is less important.

Casting silicon in a mold and then carefully controlling its rate of solidification results in an ingot that is not a single, large crystal. Instead, it consists of many regions, or grains, that are individually crystalline and which tend to have grain boundaries that run perpendicularly to the plane of the cell. Defective atomic bonds at these boundaries increase recombination and diminish current flow, resulting in cell efficiencies that tend to be a few percentage points below CZ cells. Figure 8.51 illustrates the casting, cutting, slicing, and grain boundary structure of these multicrystalline silicon (mc-Si) cells.

8.8.4 Crystalline Silicon Modules

The photovoltaic technologies described thus far result in individual thick cells that must be wired together to create modules with the desired voltage and current

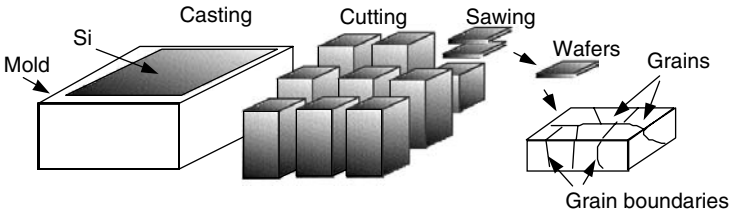


Figure 8.51 Casting, cutting and sawing of silicon results in wafers with individual grains of crystalline silicon separated by grain boundaries.

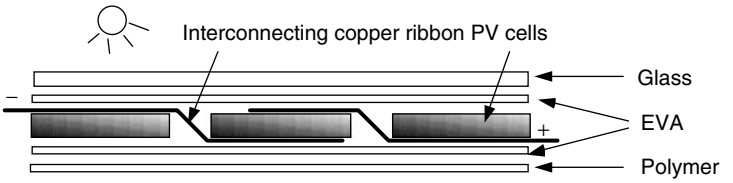


Figure 8.52 Thick crystalline cells must be wired together in series and then sandwiched in layers of glass, EVA, and polymers for structural support and weather protection.

characteristics. This wiring is done with automated soldering machines, which connect the cells in series—that is, with the front of one cell connected to the back of the next, as shown in Fig. 8.52. After soldering, the cells are laminated into a sandwich of materials that offer structural support as well as weather protection. The upper surface is tempered glass, and the cells are encapsulated in two layers of ethylene vinyl acetate (EVA). Finally, the back is covered with sheets of polymer that prevent moisture penetration.

8.9 THIN-FILM PHOTOVOLTAICS

Conventional crystalline silicon technologies (x-Si) require considerable amounts of expensive material with additional complexity and costs needed to wire individual cells together. Competing technologies, however, are based on depositing extremely thin films of photovoltaic materials onto glass or metal substrates. Thin-film devices use relatively little material (their thickness is in the micron range rather than the hundreds-of-microns range needed by crystalline silicon),* they do not require the complexity of cell interconnections, and they are particularly well suited to mass-production techniques. Their thinness allows photons that aren't absorbed to pass completely through the photovoltaic material, which offers two special opportunities. Their semitransparency means that they can be deposited onto windows, making building glass a provider of both light and

*For comparison, a human hair is roughly 100 μm in diameter.

electricity. They also lend themselves to multiple-junction, tandem cells in which photons of different wavelengths are absorbed in different layers of the device.

In exchange for these highly desirable properties, thin-film cells are not as efficient as x-Si—especially when they are not used in tandem devices. While the likelihood of significant reductions in module costs are modest for conventional crystalline silicon, many opportunities remain to increase efficiency and dramatically reduce costs using thin-film technologies. Many believe that thin films will be the dominant photovoltaic technology in the future.

8.9.1 Amorphous Silicon

Almost all of today's thin-film technology is based on amorphous (glassy) silicon (a-Si)—that is, silicon in which there is very little order to the arrangement of atoms. Since it is not crystalline, the organized tetrahedral structure in which one silicon atom bonds to its four adjacent neighbors in a precisely defined manner does not apply. While almost all of the atoms do form bonds with four other silicon atoms, there remain numerous "dangling bonds" where nothing attaches to one of the valence electrons. These dangling-bond defects act as recombination centers so that photogenerated electrons recombine with holes before they can travel very far. The key to making a-Si into a decent photovoltaic material was first discovered, somewhat by accident, in 1969, by a British team that noted a glow when silane gas SiH_4 was bombarded with a stream of electrons (Chittick et al., 1969). This led to their critical discovery that by alloying amorphous silicon with hydrogen the concentration of defects could be reduced by about three orders of magnitude. The concentration of hydrogen atoms in these alloys is roughly 1 atom in 10, so their chemical composition is approximately $Si_{0.9}H_{0.1}$. Moreover, the silicon–hydrogen alloy that results, designated as a-Si:H, is easily doped to make n-type and p-type materials for solar cells.*

The first a-Si:H solar cells were described in the literature in 1976 (Carlson and Wronski, 1976). While only 1% efficient, their potential was quickly recognized. The first commercial use of a-Si:H came in 1980 when Sanyo introduced a line of solar-powered pocket calculators. And by the mid-1980s, amorphous silicon PV modules for use outdoors were on the market. Their efficiency was only about 5% or 6% when new; within the first few months of use, this dropped to about 3% or 4%. Understanding this efficiency instability, and dealing with it, has remained a significant challenge. By the early 1990s, single-junction cells had stabilized efficiencies of about 10%, while considerably higher efficiencies have been reached for multijunction cells.

So, how can a p–n junction be formed in an amorphous material with very little organization among its atoms? Figure 8.53 shows a cross section of a simple a-Si:H cell that uses glass as the supporting superstrate. On the underside of the

*It is interesting to note that amorphous silicon is finding application in other products besides photovoltaics. The fact that it is an insulator in the dark but readily conducts current when exposed to light has led to its incorporation in photoelectronic products such as photocopiers, laser printers, and fax machines.

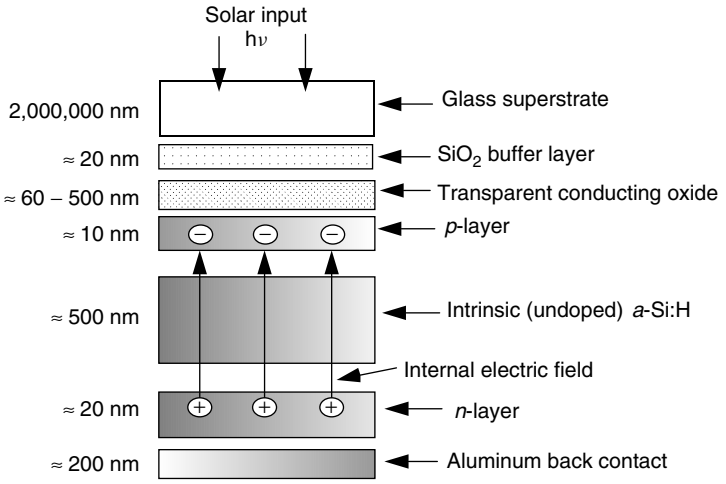


Figure 8.53 Cross section of an amorphous silicon $p-i-n$ cell. The example thicknesses are in nanometers (10^{-9} m) and are not drawn to scale.

glass a buffer layer of SiO_2 may be deposited in order to prevent subsequent layers of atoms from migrating into the glass. Next comes the electrical contact for the top of the cell, which is usually a transparent conducting oxide (TC) such as tin oxide, indium–tin oxide, or zinc oxide. The solar cell itself is often formed by subjecting gaseous silicon in the form of silane SiH_4 (or disilane Si_2H_6) to bombardment by a stream of electrons. The bombardment creates reactive silane radicals, which can attach themselves onto what becomes a growing layer of amorphous silicon. During this process, the excited silane can emit photons of light, in what is called a glow discharge. The n-layer is formed by adding phosphene gas PH_3 to the mix, while diborane gas B_2H_6 is used to form the p-layer.

The actual $p-n$ junction, whose purpose is of course to create the internal electric field in the cell to separate holes and electrons, consists of three layers consisting of the p-layer and n-layers separated by an undoped (intrinsic) region of a-Si:H As shown, the p-layer is only 10 nm thick; the intrinsic layer, or i-layer, is 500 nm; and the n-layer is 20 nm. Notice that the electric field created between the rigid positive charges in the n-layer and the rigid negative charges in the p-layer spans almost the full depth of the cell. This means that light-induced hole–electron pairs created almost anywhere within the cell will be swept across the intrinsic layer by the internal field. These amorphous silicon PVs are referred to as $p-i-n$ cells.

A variation on the a-Si $p-i-n$ cell uses a thin, flexible stainless steel substrate located at the bottom of the cell instead of the glass superstrate shown in Fig. 8.53. The stainless steel provides the mechanical strength needed while its flexibility allows modules of these cells to be literally rolled up and stored (Fig. 8.54). These flexible modules are often fitted with grommets so they can be lashed to irregular surfaces such as those found on boats.

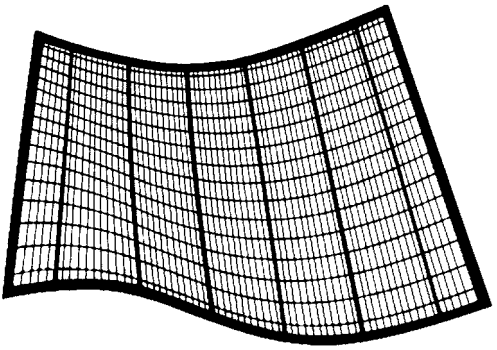


Figure 8.54 Flexible *a*-Si modules can be rolled up and stored when not in use. From SERI (1985).

The theoretical maximum efficiency for single-junction *a*-Si is 28%. The best laboratory cells today have a stabilized efficiency of just over 13 percent but commercial modules have stabilized efficiencies of only 5 to 8%.

Amorphous Silicon Processing. An important advantage of thin-film photovoltaics over conventional crystalline silicon is the ease with which they can be manufactured. Starting with rather ordinary glass, layer upon layer of materials are deposited onto the glass in a decomposition chamber. The raw materials are gaseous rather than solid, and whole modules (many cells connected in series) are created at one time rather than separately fabricating individual cells that must be soldered together to form a module.

A simplified step-by-step *a*-Si manufacturing process is described in Fig. 8.55. It begins with the deposition of a transparent conductor covering the entire underside of the glass superstrate (not shown is an SiO_2 buffer layer that might also be included). Then long narrow metal "stitch" bars, which run the length of the module, are printed onto the transparent conductor. After scribing the transparent conductor, the bars will connect the top of one cell to the bottom of the next one. Now the three layers of silicon are deposited, forming the $p-i-n$ structure of the cell itself. The final layer to be deposited is the bottom conductor, which is then scribed with long, narrow cuts that run the length of the module to complete the separation of one cell from the next. The bottom conductor may be a transparent oxide or it may be metal. If it is a transparent oxide, then photons that aren't captured in the cells can pass through the entire system, delivering some light to whatever is beneath the module.

Figure 8.56 shows how the individual cells run the entire length of the module with scribe lines that separate each cell from adjacent cells.

Multijunction or Tandem *a*-Si The band gap of *a*-Si:H is 1.75 eV, which is quite a bit higher than the 1.12 eV for crystalline silicon. Recall that higher band gaps increase voltage at the expense of lower currents (a smaller fraction of

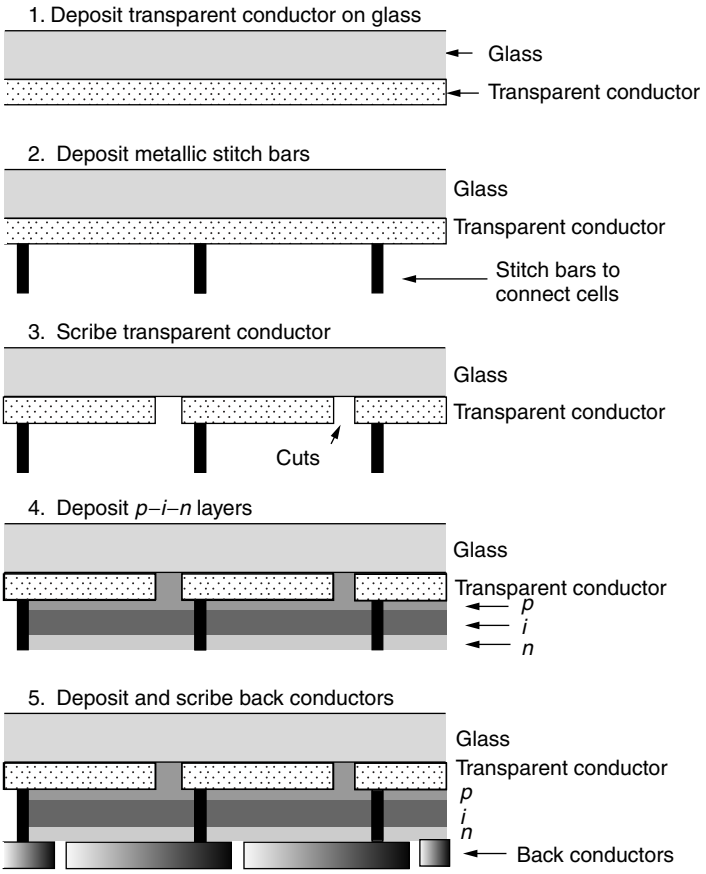


Figure 8.55 The sequence of steps taken to create a module of amorphous silicon cells.

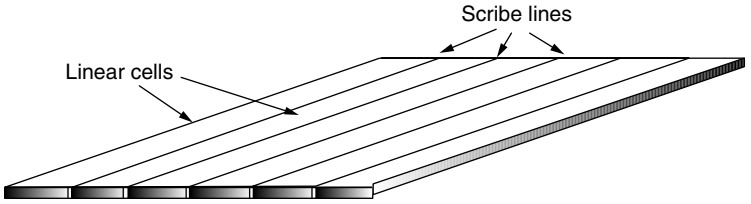


Figure 8.56 Individual cells can run the entire length of an amorphous silicon module.

solar photons have sufficient energy to create hole–electron pairs). Since power is the product of voltage and current, there will be some optimum band gap for a single-junction cell, which will theoretically result in the most efficient device. As was shown in Fig. 8.11, that optimum band gap (at AM1 corresponding to the sun directly overhead) is about 1.4 eV. Thus crystalline silicon has a band

gap somewhat too low, while a-Si has one too high to be optimum. As it turns out, however, amorphous silicon has the handy property that alloys made with other Group IV elements will cause the band gap to change. As a general rule, moving up a row in the Periodic Table increases band gap, while moving down a row decreases band gap. Referring to the portion of the Periodic Table given in Table 8.1, this rule would suggest that carbon (directly above silicon) would have a higher band gap than silicon, while germanium (directly below silicon) would have a lower band gap. To lower the 1.75-eV band gap of amorphous silicon toward the 1.4 eV optimum suggests that an alloy of silicon with the right amount of germanium (forming *a*-Si:H:Ge) can help improve cell efficiency. And that is the case.

The above discussion on *a*-Si alloys leads to an even more important opportunity, however. When *a*-Si is alloyed with carbon, for example, the band gap can be increased (to about 2 eV), and when alloyed with germanium the gap will be reduced (to about 1.3 eV). That suggests that multijunction photovoltaic devices can be fabricated by layering $p-n$ junctions of different alloys. The idea behind a multijunction cell is to create junctions with decreasing band gaps as photons penetrate deeper and deeper into the cell. As shown in Fig. 8.57a, the top junction should capture the most energetic photons while allowing photons with less energy to pass through to the next junction below, and so forth. In Fig. 8.57b, an example amorphous silicon, three-junction photovoltaic device is shown in which advantage is taken of the ability of germanium and carbon to increase or decrease the *a*-Si:H band gap.

The theoretical maximum efficiency of an ideal multijunction *a*-Si:H cell is 42%, and some estimate a practically achievable efficiency of about 24% (Carlson and Wagner, 1993). By the turn of the century, multijunction, amorphous silicon modules had a stabilized efficiency of almost 11%.

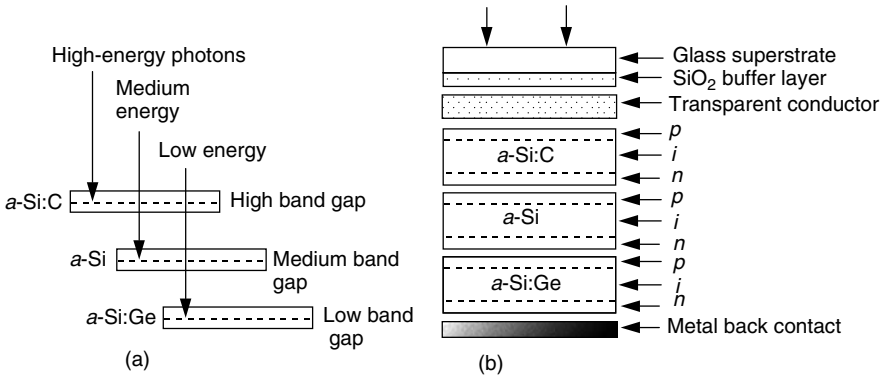


Figure 8.57 Multijunction amorphous silicon solar cells can be made by alloying *a*-Si:H (band gap ≈1.75 eV) with carbon *a*-Si:C in the top layer (≈2.0 eV) to capture the highest-energy photons and germanium *a*-Si:Ge (≈1.3 eV) in the bottom layer to capture the lowest-energy photons.

8.9.2 Gallium Arsenide and Indium Phosphide

While silicon dominates the photovoltaic industry, there is emerging competition from thin films made of compounds of two or more elements. Referring back to the portion of the Periodic Table of the elements shown in Table 8.1, recall that silicon is in the fourth column, and it is referred to as a Group IV element. These other compounds are often made up of pairs of elements from the third and fifth columns (called III–V materials), or pairs from the second and sixth columns (II–VI materials). For example, gallium, which is a Group III element, paired with arsenic, which is Group V, can be used to make gallium arsenide (GaAs) photovoltaics. Similarly, indium (Group III) and phosphorus (Group V) can be made into indium phosphide (InP) cells. Later we will consider II–VI materials such as cadmium (Group II) and tellurium (Group VI) in CdTe ("cad-telluride") cells.

Compounds such as GaAs can be grown as crystals and doped with acceptor (p-type) and donor (n-type) impurities. Common donors include Group VI elements such as Se and Te, while Group II elements such as Zn and Cd can be used as acceptors. It is even possible for elements from Group IV such as C, Si, Ge, and Sn to act as donors or acceptors, depending on which element they displace. For example, when Ge substitutes for Ga on a particular site in the lattice, it acts as a donor, but when it substitutes for As it acts as an acceptor.

As shown in Fig. 8.11, the GaAs band gap of 1.43 eV is very near the optimum value of 1.4 eV. It should not be surprising, therefore, to discover that GaAs cells are among the most efficient single-junction solar cells around. In fact, the theoretical maximum efficiency of single-junction GaAs solar cells, without solar concentration, is a very high 29%, and with concentration it is all the way up to 39% (Bube, 1998). GaAs cells with efficiencies exceeding 20% have been reported since the mid-1970s; and when used in concentrator systems in which solar energy is focused onto the cells, efficiencies approaching 30% have been realized.

In contrast to silicon cells, the efficiency of GaAs is relatively insensitive to increasing temperature, which helps them perform better than x-Si under concentrated sunlight. They are also less affected by cosmic radiation, and as thin films they are lightweight, which gives them an advantage in space applications. On the other hand, gallium is much less abundant in the earth's crust and it is a very expensive material. When coupled with the much more difficult processing required to fabricate GaAs cells, they have been too expensive for all but space applications and, potentially, for concentrator systems in which expensive cells are offset by cheap optical concentrators. Ongoing work with alloys and multijunction cells may, however, change that prognosis. Of particular interest are cells in which GaAs is coupled with other photovoltaic materials. A multijunction cell consisting of layers of GaAs and GaInP has achieved efficiencies of 29.5% for nonconcentrating AM1.5 conditions, while a hybrid, multijunction, solar concentrating cell of GaAs and Si has reached 31% efficiency.

8.9.3 Cadmium Telluride

Cadmium telluride (CdTe) is the most successful example of a II–VI photovoltaic compound. Although it can be doped in both *p*-type and *n*-type forms, it is most often used as the *p*-layer in heterojunction solar cells. Recall the distinction between homojunctions (the same material on each side of the junction) and heterojunctions (different materials). One difficulty associated with heterojunctions is the mismatch between the size of the crystalline lattice of the two materials, which leads to dangling bonds as shown in Fig. 8.58.

One way to sort out the best materials to use for the *n*-layer of CdTe cells is based on the mismatch of their lattice dimensions as expressed by their *lattice constants* (the a_1, a_2 dimensions shown in Fig. 8.58). One compound that is often used for the *n*-layer is cadmium sulfide CdS, which has a lattice mismatch of 9.7% with CdTe.

The band gap for CdTe is 1.44 eV, which puts it very close to the optimum for terrestrial cells. Thin-film laboratory cells using the *n*-CdS/*p*-CdTe heterojunction have efficiencies approaching 16% and prototype modules are reaching efficiencies beyond 9%. The equipment needed to manufacture these cells is orders of magnitude cheaper than that required for *x*-Si, and their relatively high efficiency makes them attractive candidates for mass production. While CdTe cells were used for years on pocket calculators made by Texas Instruments, full-scale modules have not yet successfully entered the marketplace.

One aspect of CdS/CdTe cells that needs to be considered carefully is the potential hazard to human health and the environment associated with cadmium. Cadmium is a very toxic substance, and it is categorized as a probable human carcinogen. Use of cadmium during the manufacture of CdTe cells needs to be carefully monitored and controlled to protect worker health, but apparently necessary safety precautions are relatively straightforward. Waste cadmium produced during the manufacturing process needs to be kept out of the environment and

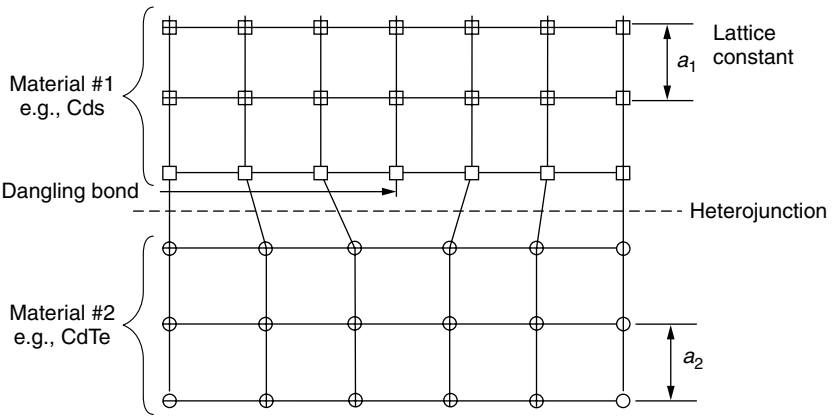


Figure 8.58 The mismatch between heterojunction materials leads to dangling bonds as shown.

should be recycled. The question then arises as to what precautions are necessary once modules have been manufactured and installed. CdS/CdTe modules contain about 6 g of cadmium per square meter of surface area, but it is completely sealed inside of the module so it should pose no risk under normal circumstances. If all of the cadmium in a rooftop PV system were to vaporize in a fire, however, and be inhaled by an individual, it would be pose a very serious health risk. But the likelihood of someone inhaling enough cadmium to cause harm without also having inhaled a lethal dose of smoke is considered to be insignificantly small.

8.9.4 Copper Indium Diselenide (CIS)

The goal in exploring compounds made up of a number of elements is to find combinations with band gaps that approach the optimum value while minimizing inefficiencies associated with lattice mismatch. Copper indium diselenide, $CuInSe_2$, better known as "CIS," is a *ternary* compound consisting of one element, copper, from the first column of the Periodic Table, another from the third column, indium, along with selenium from the sixth column. It is therefore referred to as a I–III–VI material. A simplistic way to think about this complexity is to imagine that the average properties of Cu (Group I) and In (Group III) are somewhat like those of an element from the second column (Group II), so the whole molecule might be similar to a II–VI compound such as CdTe (Bube, 1998).

While the crystal structure of silicon is a simple tetrahedral that is easy to understand and visualize, crystalline $CuInSe_2$ is much more complicated. As shown in Fig. 8.59, each selenium atom serves as the center of a tetrahedron of

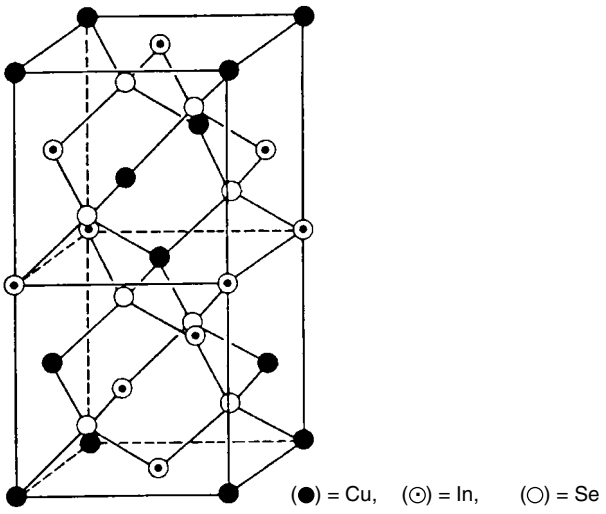


Figure 8.59 The crystalline structure of $CuInSe_2$ or "CIS." From Bube (1998) based on Kazmerski and Wagner (1985).

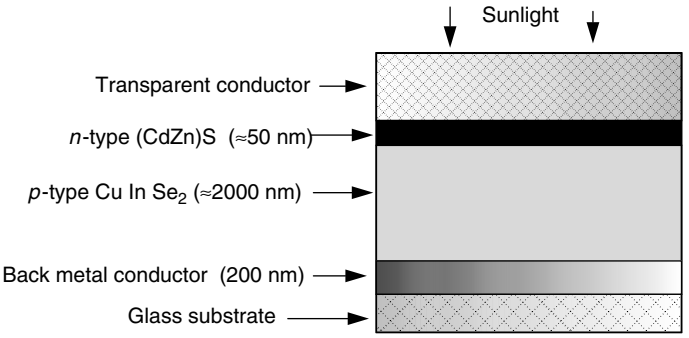


Figure 8.60 Structure of a simple, thin-film copper indium diselenide (CIS) cell.

two Cu and two In atoms, and each Cu atom is the center of a tetrahedron of selenium atoms.

CIS cells often use a thin layer of cadmium and zinc sulfide (CdZn)S for the *n*-layer as shown in Fig. 8.60.

With the substitution of gallium for some of the indium in the CIS material, the relatively low 1.04-eV band gap of CIS is increased and efficiency is improved. This is consistent with our interpretation of the Periodic Table in which band gap increases for elements in higher rows of the table (Ga is above In). The resulting $CuIn_{1-x}Ga_xSe_2$ alloy is called "CIGS" for short. By 2003, laboratory CIGS cells had achieved efficiencies of almost 20% and production modules had efficiencies in the range of 8 to 10%.

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PROBLEMS

8.1 For the following materials, determine the maximum wavelength of solar energy capable of creating hole-electron pairs:

a. Gallium arsenide, GaAs, band gap 1.42 eV.

b. Copper indium diselenide, $CuInSe_2$, band gap 1.01 eV

c. Cadmium sulfide, CdS, band gap 2.42 eV.

8.2 A *p-n* junction diode at 25°C carries a current of 100 mA when the diode voltage is 0.5 V. What is the reverse saturation current, I_0?

8.3 For the simple equivalent circuit for a 0.005 m^2 photovoltaic cell shown below, the reverse saturation current is $I_0 = 10^{-9}$ A and at an insolation of 1-sun the short-circuit current is $I_{SC} = 1$ A,. At 25°C, find the following:

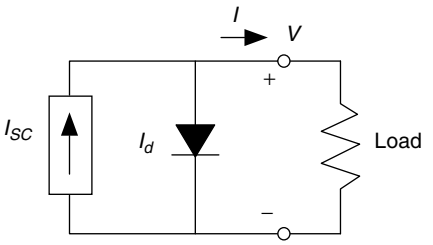


Figure P8.3

a. The open-circuit voltage.

b. The load current when the output voltage is $V = 0.5$ V.

c. The power delivered to the load when the output voltage is 0.5 V.

d. The efficiency of the cell at $V = 0.5$ V.

8.4 The equivalent circuit for a PV cell includes a parallel resistance of $R_P = 10\ \Omega$. The cell has area 0.005 m^2, reverse saturation current of $I_0 = 10^{-9}$ A and at an insolation of 1-sun the short-circuit current is $I_{SC} = 1$ A, At 25°C, with an output voltage of 0.5 V, find the following:

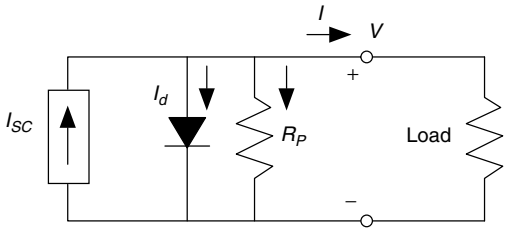


Figure P8.4

a. The load current.

b. The power delivered to the load.

c. The efficiency of the cell.

8.5 The following figure shows two I-V curves. One is for a PV cell with an equivalent circuit having an infinite parallel resistance (and no series resistance). What is the parallel resistance in the equivalent circuit of the other cell?

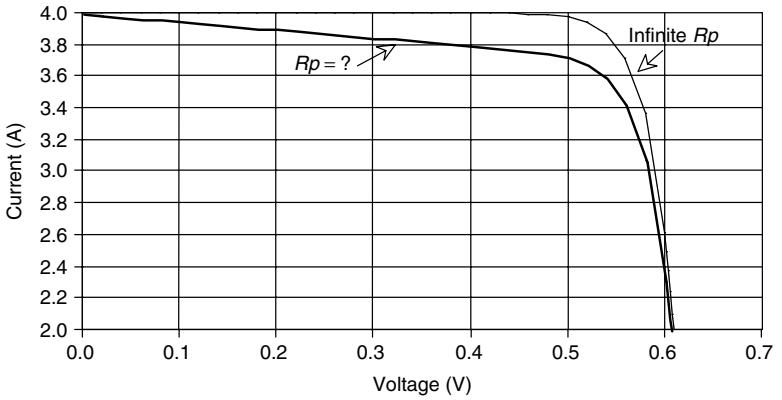


Figure P8.5

8.6 The following figure shows two I-V curves. One is for a PV cell with an equivalent circuit having no series resistance (and infinite parallel resistance). What is the series resistance in the equivalent circuit of the other cell?

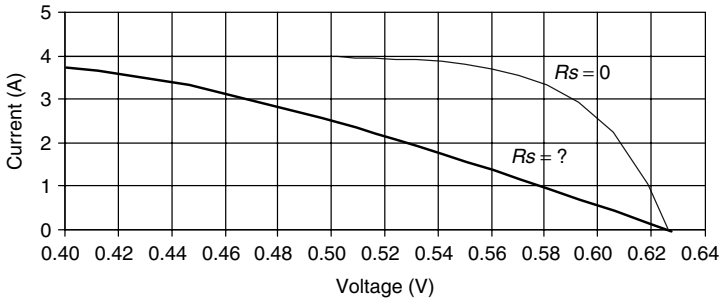


Figure P8.6

8.7 Estimate the cell temperature and power delivered by a 100-W PV module with the following conditions. Assume 0.5%/°C power loss.

a. NOCT = 50°C, ambient temperature of 25°C, insolation of 1-sun.

b. NOCT = 45°C, ambient temperature of 0°C, insolation of 500 W/m^2.

c. NOCT = 45°C, ambient temperature of 30°C, insolation of 800 W/m^2.

8.8 A module with 40 cells has an idealized, rectangular I-V curve with I_{SC} = 4 A and V_{OC} = 20 V. If a single cell has a parallel resistance of 5 Ω and negligible series resistance, draw the I-V curve if one cell is completely shaded. What current would it deliver to a 12-V battery (vertical I-V load at 12 V)?

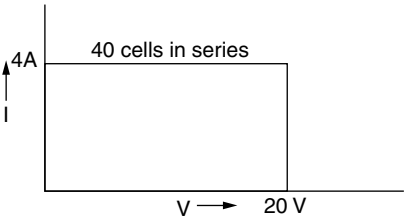


Figure P8.8

8.9 Suppose a PV module has the 1-sun I-V curve shown below. Within the module itself, the manufacturer has provided a pair of bypass diodes to help the panel deliver some power even when many of the cells are shaded. Each diode bypasses half of the cells, as shown. You may consider the diodes to be "ideal;" that is, they have no voltage drop across them when conducting.

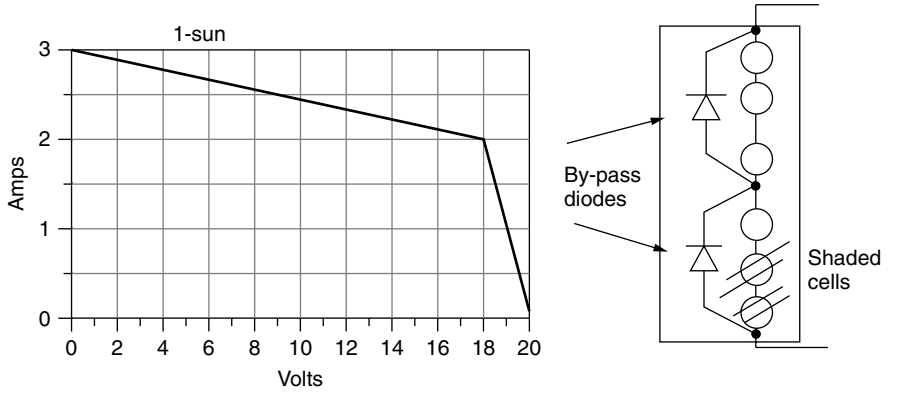


Figure P8.9

Suppose there is enough shading on the bottom cells to cause the lower diode to start conducting. Draw the new "shaded" I-V curve for the module.

CHAPTER 9

PHOTOVOLTAIC SYSTEMS

9.1 INTRODUCTION TO THE MAJOR PHOTOVOLTAIC SYSTEM TYPES

The focus of this chapter is on the analysis and design of photovoltaic (PV) systems in their three most commonly encountered configurations: systems that feed power directly into the utility grid, stand-alone systems that charge batteries, perhaps with generator back-up, and applications in which the load is directly connected to the PVs as is the case for most water-pumping systems.

Figure 9.1 shows a simplified diagram of the first of these systems—a grid-connected or *utility interactive* (UI) system in which PVs are supplying power to a building. The PV array may be pole-mounted, or attached externally to the roof, or it may become an integral part of the skin of the building itself. PV roofing shingles and thin-film PVs applied to glazing serve dual purposes, power, and building structure, and when that is the case the system is referred to as *building-integrated photovoltaics* (BIPV).

The photovoltaics in a grid-connected system deliver dc power to a power conditioning unit (PCU) that converts dc to ac and sends power to the building. If the PVs supply less than the immediate demand of the building, the PCU draws supplementary power from the utility grid, so demand is always satisfied. If, at any moment, the PVs supply more power than is needed, the excess is sent back onto the grid, potentially spinning the electric meter backwards. The system is relatively simple since failure-prone batteries are not needed for back-up power, although sometimes they may be included if utility outages are problematic. The

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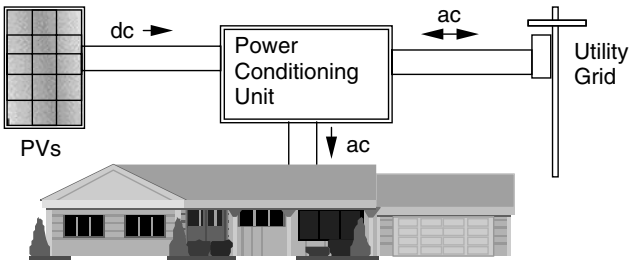


Figure 9.1 Simplified grid-connected PV system.

power-conditioning unit also helps keep the PVs operating at the most efficient point on their $I-V$ curves as conditions change.

Grid-connected PV systems have a number of desirable attributes. Their relative simplicity can result in high reliability; their maximum-power-tracking unit assures high PV efficiency; their potential to be integrated into the structure of the building means that there are no additional costs for land and, in some cases, materials displaced by PVs in such systems may offset some of their costs; and finally, their ability to deliver power during the middle of the day, when utility rates are highest, increases the economic value of their kilowatt-hours. All of these attributes contribute to the cost effectiveness of these systems. On the other hand, they have to compete with the relatively low price of utility power.

Figure 9.2 shows the second system, which is an off-grid, stand-alone system with battery storage and a generator for back-up power. In this particular system, an inverter converts battery dc voltages into ac for conventional household electricity, but in very simple systems everything may be run on dc and no inverter may be necessary. The charging function of the inverter allows the generator to top up the batteries when solar is insufficient.

Stand-alone PV systems can be very cost effective in remote locations where the only alternatives may be noisy, high-maintenance generators burning relatively expensive fuel, or extending the existing utility grid to the site, which can cost thousands of dollars per mile. These systems suffer from several inefficiencies, however, including battery losses and the fact that the PVs usually operate well off of the their most efficient operating point. Moreover, inefficiencies are often increased by mounting the array at an overly steep tilt angle to supply relatively uniform amounts of energy through the seasons, rather than picking an angle that results in the maximum possible annual energy delivery. These systems also require much more attention and care than stand-alone systems; and if generator usage is to be minimized (or eliminated), those using the energy may need to modify their lifestyles to accommodate the uneven availability of power as the seasons change or the weather deteriorates.

The third system type that we will pay close attention to has photovoltaics directly coupled to their loads, without any batteries or major power conditioning equipment. The most common example is PV water pumping in which the wires from the array are connected directly to the motor running a pump (Fig. 9.3).